

Unit-2Diffraction (7)

→ The process of bending of light from the corners of an obstacle which is comparable with wavelength of light and spreading into the geometrical shadow is known as diffraction. There are mainly two types of diffraction.

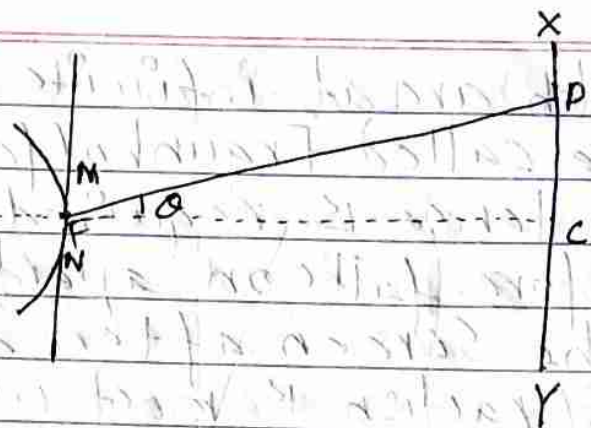
1. Fresnel Diffraction

The diffraction in which the source of light or screen or both are at finite distance from the aperture called Fresnel diffraction. In this diffraction no lenses are used to make the rays parallel or converge. The incident wavefront is either spherical or cylindrical.

Fresnel Assumption

To find the resultant effect at external point due to a wavefront will depend upon following factors.

Consider S is a point source of monochromatic light, MN is small aperture and XY is the screen as shown in figure. To find resultant amplitude at a point P on the screen Fresnel assumed following assumption:



Fig

- 1) A wavefront can be divided into a large number of small elements or zones is called fresnel zones. Each of small area. The resultant effect at any point 'P' on the screen will depends on the combined effect of all the secondary waves originating from various zones.
- 2) The effect at any point due to particular zone will depend on the distance of the point from the zone.
- 3) The effect at point 'P' will also depends upon the obliquity (θ) of the point with respect to zone. The obliquity factor is proportional to $(1 + \cos \theta)$ where θ is angle between FC and FP considering wavefront at point F. The effect will be maximum if $\theta = 0$. As the θ increase the resultant effect decreases. For $\theta = 180^\circ$ resultant effect is 0 which explains that non-existence of wave in backward direction. An application of Fresnel diffraction is in the construction of zone plate.

* Fraunhofer Diffraction

The diffraction in which the source of light,

screen or both are at infinite distance from the aperture called Fraunhofer diffraction. The convex lens is required to make light parallel before falls on aperture and to focus on the screen after diffraction. This type of diffraction is most used in case of plane transmission grating. In this diffraction, incident wavefront is plane.

Zone Plate

It is a transparent plate on which concentric circles whose radii are proportional to square root of natural number. The alternative zones are painted black to obstruct the light. Such a plate behaves like convex lens and produces image of a source of light on the screen placed at suitable distance.

Construction:

To construct a zone plate concentric circles are drawn on white paper such that the radii are proportional to square root of natural number. The odd numbered zones are covered with black ink and reduced photograph is taken and it appears as given in figure B. Hence, the odd zones are transparent to incident light and even zones will cut off light.

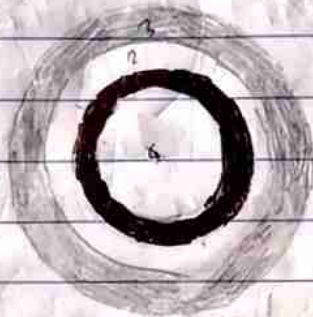


Fig (a): Concentric circle on paper

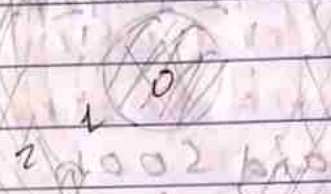
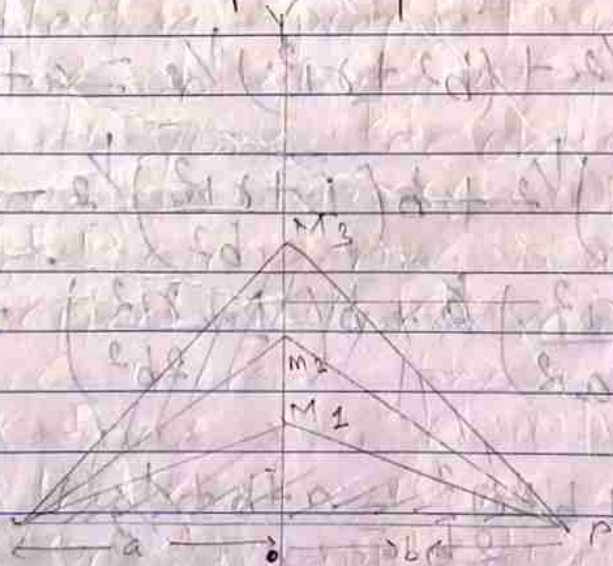


Fig (b): Reduced Photograph.

Suppose XY represents the section of zone plate held $\perp$ to plane of paper and S is point source of light. Suppose P be the position of screen for a bright image, A is the distance of source from zone plate and B is the distance of screen from plate as shown in figure



Suppose OM_1 , OM_2 , OM_3 (r_1, r_2, r_3) are the radii of first, second and third half period zones and the position of screen is such that from one zone to next there is an increase

in path difference $\lambda/2$. Now, we have
 $SO + OP = a + b$

$$SM_1 + M_1P = a + b + \lambda/2 \quad \text{--- (1)}$$

$$SM_2 + M_2P = a + b + \lambda$$

and soon

From figure In $\triangle SM_1O$ we have

$$SM_1 = (SO^2 + OM_1^2)^{1/2}$$

$$= (a^2 + r_1^2)^{1/2}$$

$$\therefore SM_1 = (a^2 + r_1^2)^{1/2}$$

Similarly In $\triangle OM_1P$ we have

$$PM_1 = (OP^2 + OM_1^2)^{1/2}$$

$$\therefore PM_1 = (b^2 + r_1^2)^{1/2}$$

Now, from eqn (1) we have

$$(a^2 + r_1^2)^{1/2} + (b^2 + r_1^2)^{1/2} = a + b + \lambda/2$$

$$\text{or, } a \left(\frac{1 + \frac{r_1^2}{a^2}}{a^2} \right)^{1/2} + b \left(\frac{1 + \frac{r_1^2}{b^2}}{b^2} \right)^{1/2} = a + b + \lambda/2$$

$$\text{or, } a \left(\frac{1 + \frac{r_1^2}{a^2}}{2a^2} \right) + b \left(\frac{1 + \frac{r_1^2}{b^2}}{2b^2} \right) = a + b + \lambda/2$$

$$\text{or, } a + \frac{r_1^2}{2a} + b + \frac{r_1^2}{2b} = a + b + \lambda/2$$

$$\text{or, } \frac{r_1^2}{2a} + \frac{r_1^2}{2b} = \frac{\lambda}{2}$$

$$\frac{r_1^2}{2} \left(\frac{1}{a} + \frac{1}{b} \right) = \frac{\lambda}{2}$$

$$n^2 \left(\frac{1}{a} + \frac{1}{b} \right) = 1 \quad \text{--- (2)}$$

Similarly for n th zone the relation can be obtained

$$r_n^2 \left(\frac{1}{a} + \frac{1}{b} \right) = n\lambda$$

$$\therefore \frac{1}{a} + \frac{1}{b} = \frac{n\lambda}{r_n^2}$$

$$\therefore \frac{1}{a} + \frac{1}{b} = \frac{1}{fh}$$

where $fh = \frac{r_n^2}{n\lambda}$ is the focal length of the n th zone.

This shows that zone plate acts as a converging lens. A zone plate has number of foci which depends on number of zones.

Similar Between Zone Plate And Convex lens

1. Both zone plate and convex lens form a real image of the object.
2. The focal length of both depends upon wave length λ and suffers chromatic aberration.

Differences Between Zone Plate And Convex lens

- | Zone plate | convex lens |
|---|---|
| 1. In zone plate image is formed by diffraction phenomenon. | 1. In convex lens image is formed by refraction of light. |
| 2. In zone plate it has multiple foci. | 2. It has single focus. |
| 3. Waves reaching image point through any two alternate zones differ in path by λ . | 3. All the rays reaching image point has zero path difference. |
| 4. Chromatic aberration is more severe than in convex lens. | 4. Chromatic aberration is not severe. |
| 5. The focal length $fh = \frac{r_n^2}{n\lambda}$. | 5. For convex lens $\frac{1}{f} = n-1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$ |

Fraunhofer Diffraction At Single Slit

[IMP]

Consider a parallel beam of light incident normally on a slit AB of width ' a ' as shown in figure. After passing through the slit the beam is focused on the screen XY by means of lens L_2 .

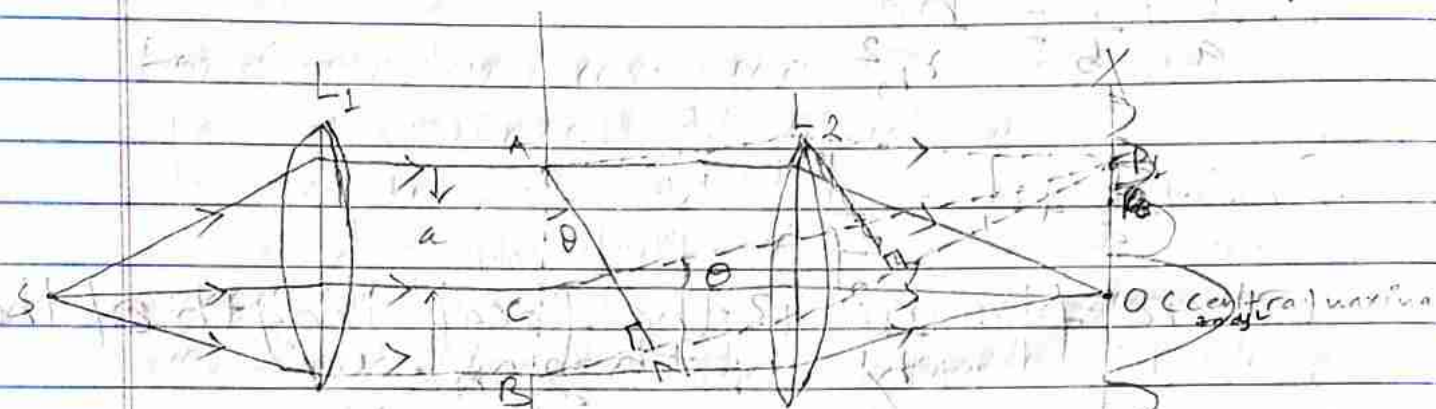


fig: Fraunhofer Diffraction At single Slit

When a plane wavefront is incident on slit AB each point on the plane wavefront AB act as the source of secondary wavelets. These wavelets initially in same phase and spread out in all direction the combined effect at any point on the screen can be found by the superposition of these wavelets.

Consider O as a point at the centre of screen. All the waves in the direction of incident light travel equal path while reaching at point O. Thus, central bright fringe of maximum intensity is observed at point 'O' but on either side of O alternative dark and bright fringes of decreasing intensity are observed.

Suppose P_1 be the point on the screen at which wavelets travelling in a direction by making an angle ' θ ' with a CO and focused by means of lens L_2 . The wavelets from different part of slit will not reach at point ' P_1 ' in same phase because they have unequal optical path. The point P_1 is maxima or minima that depends upon path difference which is given by BN on the figure

In rt-angled $\triangle ABN$

$$\sin \theta = \frac{BN}{AB}$$

AB

$$BN = AB \sin \theta$$

$$\therefore BN = a \sin \theta$$

$\therefore$ path difference

$$BP_1 - AP_1 = BN = a \sin \theta$$

Suppose a path difference between the secondary waves originating from A and B is equal to the wavelength of light. To find resultant amplitude at point P_1 let us divide the slit AB into equal halves i.e. $AC = BC = \frac{a}{2}$. The path difference between secondary waves starting from A and C equal to $\frac{1}{2}$ and similarly path difference between waves starting from C and B equal to $\frac{1}{2}$. So, there is phase difference of 180° . Hence they form minima at point P_1 .

$\therefore$ For first Minima

We have

$$\frac{a \sin \theta}{2} = \frac{\lambda}{2}$$

$$a \sin \theta = \lambda$$

In general for n th secondary minima

$$a \sin \theta_n = n\lambda \quad \text{--- (1)}$$

For Secondary maxima

path difference = odd integral multiple of $\frac{\lambda}{2}$

$$i.e. a \sin \theta_n = (2n+1) \frac{\lambda}{2}$$

Where $n = 1, 2, 3$

Thus, we find that the diffraction pattern due to single slit consist of central maxima at 0 followed by secondary maxima and minima on either side of central maxima. The intensity distribution of diffraction pattern due to single slit as a function of θ (θ) is given below

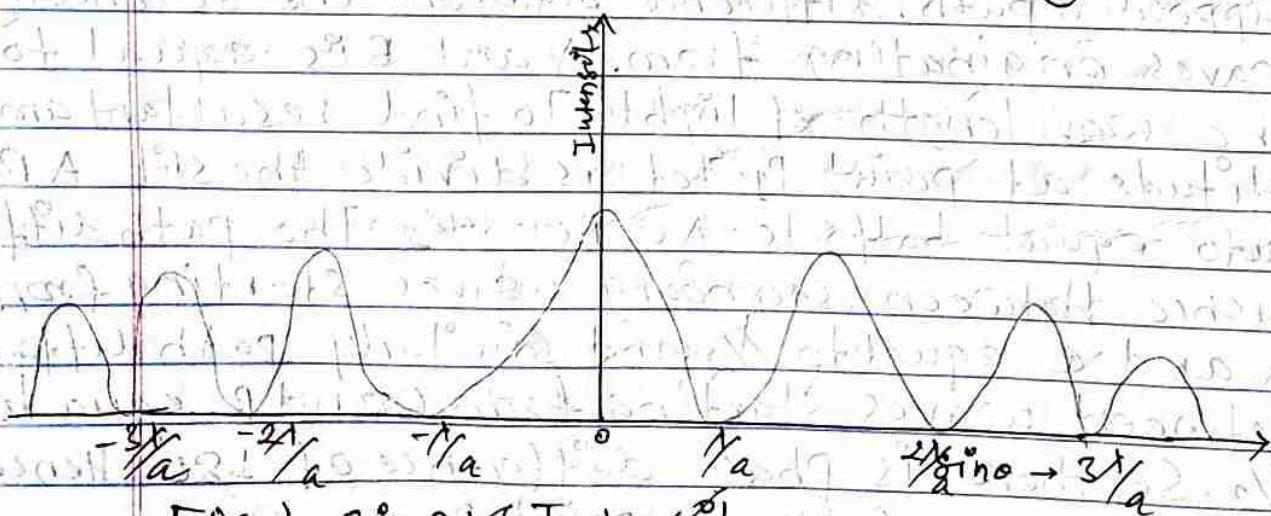


Fig: $\sin \theta$ vs Intensity

Width of Central Maxima

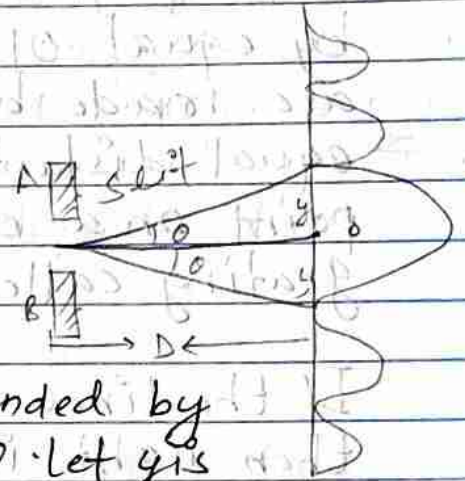
It means that the distance between first minima on either side of central maxima. We have for first minima

$$a \sin \theta = \lambda$$

$$\sin \theta = \frac{\lambda}{a}$$

For small angle $\sin \theta \approx \theta$

$$\theta = \frac{\lambda}{a} \quad \text{--- (1)}$$



Here, in figure the angle subtended by whole central maxima is 2θ . Let y is the distance of point 'o' from first minima then width of central maxima is $2y$. If D is the distance between slit and screen then, from figure,

$$\tan \theta = \frac{y}{D}$$

for small angle $\tan \theta \approx \theta$

$$\theta = \frac{y}{D} \quad \text{--- (2)}$$

Now, from (1) and (2) we have

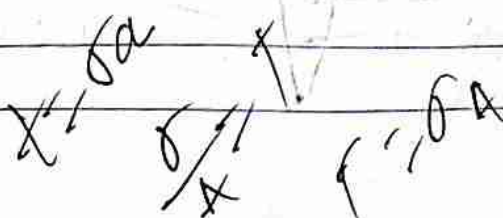
$$\frac{y}{D} = \frac{\lambda}{a}$$

$$\therefore y = \frac{\lambda D}{a}$$

but width of central maxima is $2y$

$$\therefore \text{width of central maxima } (\beta) = 2y$$

$$\therefore \beta = \frac{2\lambda D}{a}$$



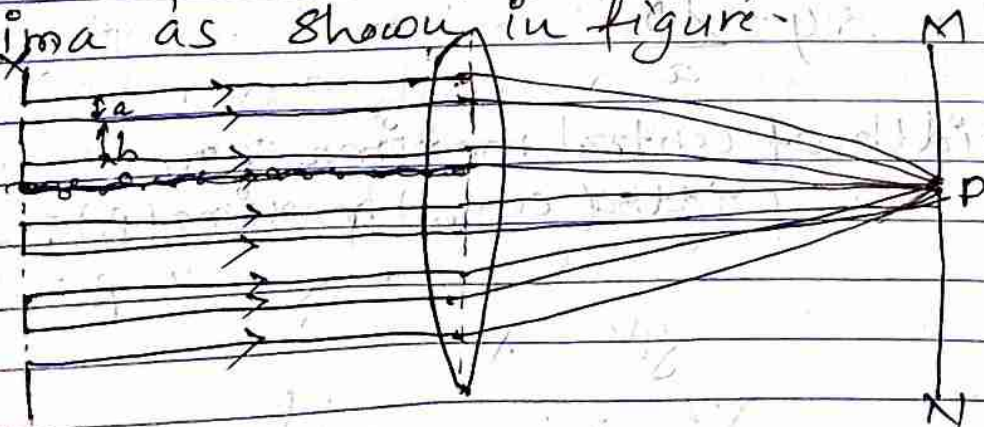
Plane Transmission Grating

A plane transmission grating is an arrangement which consists of large number of parallel slits of equal width separated from one another by equal opaque spaces. Such opaque spaces are made by ruling a large number of fine equal distant and parallel lines with a diamond point on an optically glass plate. Such type of grating called plane transmission grating.

If the lines are drawn on silvered surface plane then light is reflected from the position of silvered surfaces between any two line. Such type of surface act as reflection grating.

Theory of Plane Transmission Grating

Let XY is the grating surface and MN is the screen both are perpendicular to plane of paper. Suppose width of each slit is ' a ' and width of opaque space is ' b '. When a plane wavefront be incident on a grating surface then all the secondary waves travelling in same direction and same phase come to focus at point ' P ' on a screen which corresponds to position of central bright maxima as shown in figure.



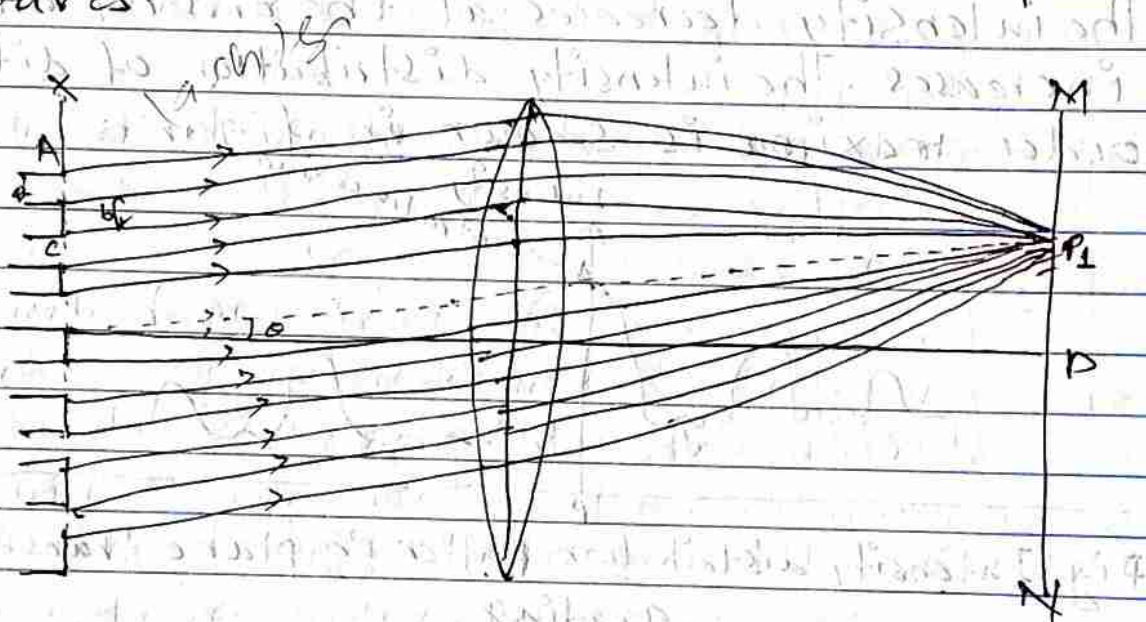


(a+b) sin θ = $n\lambda$

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Now consider the secondary wave travelling in the direction inclined at angle θ with the direction of incident light as shown in figure. These secondary wave focus at point P_1 on the screen by placing the lense to the direction of secondary waves.



The intensity at P_1 will depends on path difference between the secondary wave originating from point A and C of neighbouring slit. The path difference between waves is equal to $(a+b) \sin \theta$. The point P_1 will be maximum if path difference is equal to integral multiple of λ . Therefore for n th secondary maxima.

$$\therefore (a+b) \sin \theta_n = n\lambda \quad \text{--- (1)}$$

where, $a+b$ = grating element which is determined from the number of ruling per inch on the grating. If no. of ruling is N in 1 inch then we have

$$N(a+b) = 2.54 \text{ cm}$$

$$a+b = \frac{2.54}{N} \text{ cm}$$

N

$$a+b = \frac{2.54}{N}$$

N

From eqn (1) we have

For first order of maxima the angle of diffraction is

$$\sin \theta_1 = \frac{\lambda}{a+b}$$

The intensity decreases as the order of maxima increases. The intensity distribution of different order maxima is shown in figure.

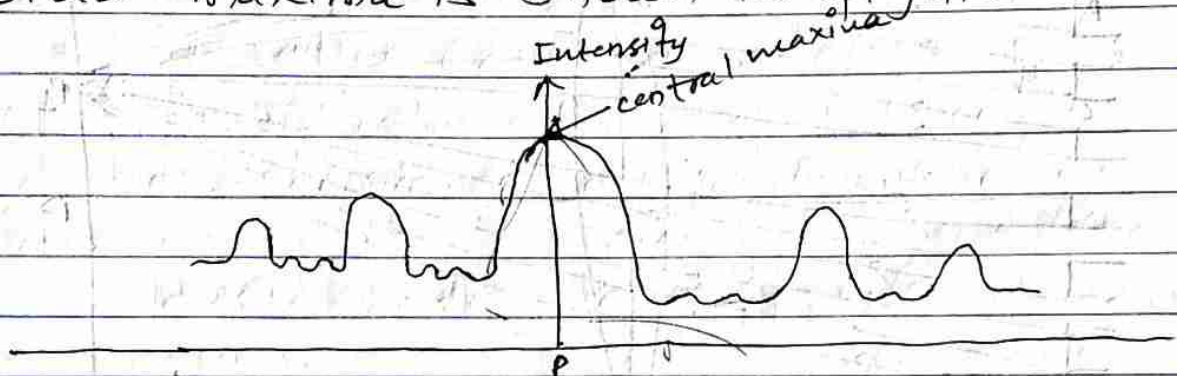


Fig: Intensity distribution pattern in plane transmission grating

Dispersive Power of Grating.

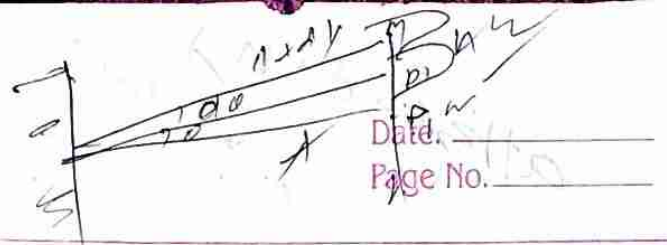
It is defined as the ratio of difference in the angle of diffraction of any two neighbouring spectral lines to the difference in wave length between any two spectral lines. We have n th order maxima

$$(a+b) \sin \theta_n = n\lambda$$

Differentiating w.r.t. θ we get

$$(a+b) \cos \theta \, d\theta = n \, d\lambda$$

$$\therefore \frac{d\theta}{d\lambda} = \frac{n}{(a+b) \cos \theta}$$



d is the no. of lines per cm of grating surface then we have

$$\frac{d\theta}{d\lambda} = \frac{N}{(a+b) \cos \theta}$$

Resolving Power of Optical Instrument.

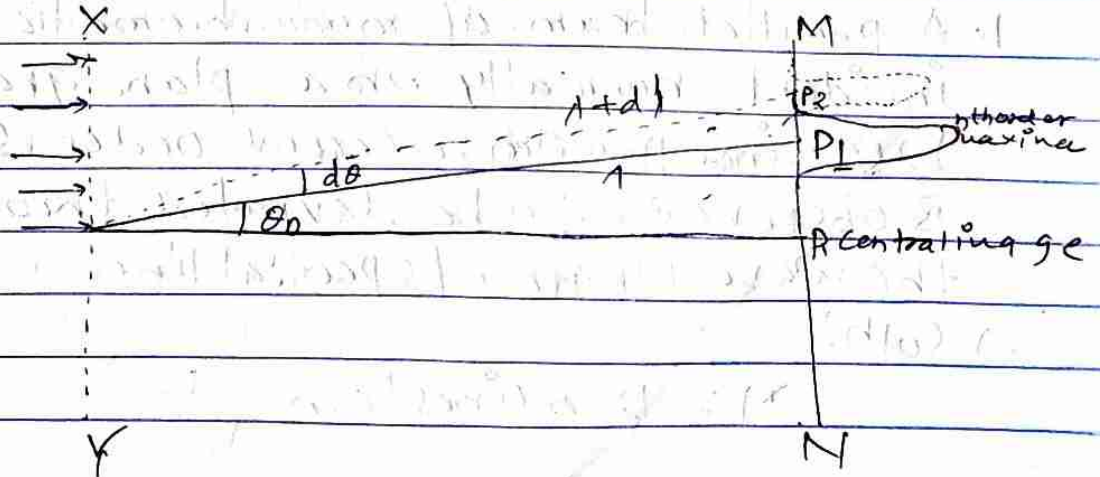
The ability of an optical instrument to resolve the images of two nearly by object is called resolving power.

> Resolving Power of Plane Transmission Grating

The resolving power of a grating is defined as the ratio of wavelength of any spectral line to the difference in wavelength between this line and neighbouring line such that these two lines are just appears to be resolved.

i.e. Resolving power of grating = $\frac{\lambda}{d\lambda}$

Let XY is the grating surface and MN is the field view of Telescope. Suppose P_1 is the point of n^{th} primary maxima of spectral line of wave length λ at an angle of diffraction θ_n and P_2 is the position of n^{th} primary maxima of second spectral line of wave length $\lambda + d\lambda$ at a angle of diffraction $\theta_n + d\theta$ as shown in figure.



at θ_n $\sin \theta_n = n\lambda$
 at $\theta_n + d\theta$ $\sin(\theta_n + d\theta) = n(\lambda + d\lambda)$

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For n th primary maxima of wavelength λ
 $(a+b) \sin \theta_n = n\lambda$ - (1)

Similarly n th primary maxima of second spectral line of wave length $\lambda + d\lambda$
 $(a+b) \sin(\theta_n + d\theta) = n(\lambda + d\lambda)$ - (2)

The two lines are just just resolved then, for angle of diffraction $\theta_n + d\theta$ an additional path difference of $\frac{\lambda}{N}$ is introduced where N is the total number of lines per cm on the grating surface. Then we have

$$(a+b) \sin(\theta_n + d\theta) = n\lambda + \frac{\lambda}{N}$$
 - (3)

Equation eqn (2) and (3) we get

$$n(\lambda + d\lambda) = n\lambda + \frac{\lambda}{N}$$

$$n\lambda + nd\lambda = n\lambda + \frac{\lambda}{N}$$

$$nd\lambda = \frac{\lambda}{N}$$

$$\frac{d\lambda}{\lambda} = \frac{1}{Nn}$$

N = no. of lines per cm
 n = order

1. A parallel beam of monochromatic B allowed to incident normally on a plane grating having 1250 lines per cm a second order spectral line B observed to be deviated through 30° . calculate the wave length of spectral line

→ Soln:

$$N = 1250 \text{ lines/cm} \quad \text{or} \quad 1250 \frac{\text{lines}}{\text{cm}}$$

$$\therefore a+b = \frac{1}{N} = \frac{1}{1250} \text{ cm}$$

$$h=2$$

Now,

$$a+b \sin \theta = h\lambda$$

$$\frac{1}{1250} \times \sin 30^\circ = 2\lambda$$

$$\therefore \lambda = 2 \times 10^{-6} \text{ m}$$

2. Light is incident normally on a grating 0.5 cm wide with 2500 lines. find angle of diffraction for principle maxima of two sodium line in the first order spectrum $\lambda_1 = 5890 \text{ \AA}$ and $\lambda_2 = 5896 \text{ \AA}$. Also, calculate their resolution. Are two line resolved.

→ solution:

$$N = \frac{2500}{0.5} \text{ lines per cm}$$

$$a+b = \frac{1}{N}$$

$$d = a+b = \frac{1}{5000}$$

Now

$$a+b \sin \theta_1 = h\lambda_2$$

$$\frac{1}{5000} \sin \theta_1 = 1 \times \lambda_2$$

$$\frac{1}{5000} \sin \theta_1 = 1 \times 5890 \times 10^{-8}$$

$$\therefore \theta_1 = 2.1687^\circ \approx 17.12^\circ$$

Again,

$$(a+b) \sin \theta_1' = n \lambda$$

$$\frac{1}{5000} \sin \theta_1' = 1 \times 5896 \times 10^{-8}$$

$$\theta_1' = 17.14^\circ$$

Then,

$$\frac{\Delta d}{d} = \frac{n \lambda}{d}$$

Unit-VII

Optical Fibre (8)

✓✓ Optical fibre is a very thin glass or plastic cond. w/ (pipe) designed to guide light waves along the length of the fibre with the help of successive total internal reflections from the side walls of the fibre.

➤ Construction:

Optical fibre consists of a thin glass core of refractive index μ_1 about $(5-50 \mu\text{m})$ micrometer $(\times 10^6)$ in diameter surrounded by a glass coating called cladding of refractive index μ_2 such that $(\mu_1 > \mu_2)$ which is enclosed in a protective plastic jacket called sheath. The sheath protects the cladding and the core from abrasion, contamination and harmful influence of moisture in addition it increase the mechanical strength of fibre.

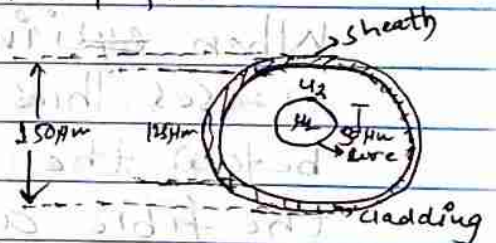


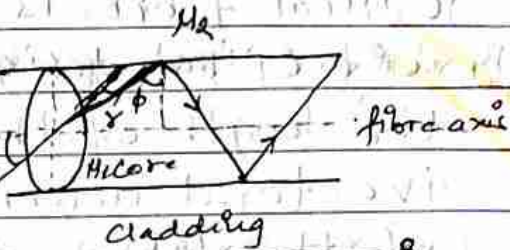
Fig: Optical fibre

Acceptance Angle.

✓✓ It is defined as the maximum angle of incidence that a light wave make with the axis of fibre so that it may propagate through optical fibre.

Suppose AB is optical fibre through which light being sent. The end at which light enters

is called launching end. Let μ_1 is refractive index of core and that of cladding is μ_2 such that ($\mu_1 > \mu_2$). Let refractive index of medium from which light enters the fibre is μ_a .



Consider a ray of light incident on a fibre at an angle of incidence i . This ray get refracted at angle r and strikes the core cladding interface at an angle ϕ . If ϕ is equal to or greater than critical angle between core and cladding. This ray suffers total internal reflection.

When i increases r also increases then ϕ decreases. Thus as i increases ϕ may fall below the critical value such ray refracts into the fibre and reaches core cladding interface at an angle less than critical angle. Hence it travels into the optical fibre and emerges out of fibre into air.

$\therefore$ Maximum value of i occurs when $i = c$

Now, we have

$$\mu_1 = \frac{\sin i}{\sin r}$$

$$\mu_2 = \frac{\sin i}{\sin r_c}$$

$$\frac{\mu_1}{\mu_2} = \frac{\sin i}{\sin r}$$

$$\sin r = \sin(90^\circ - \phi)$$

$$\sin r = \cos \phi$$

$$\therefore \sin i = \left(\frac{\mu_1}{\mu_2}\right) \sin r$$

$$\sin i = \left(\frac{\mu_1}{\mu_2}\right) \cos \phi$$

When $\phi = c$

$$\sin(i_{\max}) = \left(\frac{\mu_1}{\mu_2}\right) \cos c$$

But we have

$$\sin c = \frac{\mu_2}{\mu_1}$$

$$\therefore \cos c = \sqrt{1 - \sin^2 c}$$

$$= \sqrt{1 - \frac{\mu_2^2}{\mu_1^2}}$$

$$\therefore \sin(i_{\max}) = \frac{\mu_1}{\mu_2} \sqrt{1 - \frac{\mu_2^2}{\mu_1^2}}$$

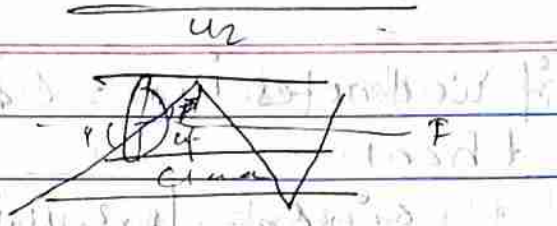
$$= \frac{\mu_1}{\mu_2} \sqrt{\frac{\mu_1^2 - \mu_2^2}{\mu_1^2}}$$

$$= \sqrt{\mu_1^2 - \mu_2^2}$$

This is expression for acceptance angle.

If the medium from which light enters the fibre is air $\mu_a = 1$

$$\therefore \sin i_{\max} = \sqrt{\mu_1^2 - \mu_2^2}$$



if we denotes $i_{\max} = \theta_a$
then

$$\sin \theta_a = \sqrt{\mu_1^2 - \mu_2^2} \quad - (2)$$

Which is expression for acceptance angle.

Numerical Aperture

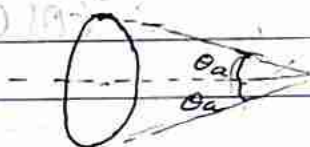
It is defined as the ~~sign~~ of the acceptance angle.

$$\therefore \text{Numerical aperture } \sin \theta_a = \sqrt{\mu_1^2 - \mu_2^2}$$

Numerical aperture is a factor on which depends the light gathering ability of the fibre.

Acceptance Cone

The light ray contained in cone having vertex angle $2\theta_a$ are all accepted (propagated) and transmitted through fibre.



Relative Refractive Index Difference

We have $\mu_1^2 - \mu_2^2 = (\mu_1 + \mu_2)(\mu_1 - \mu_2)$

$$= \frac{\mu_1 + \mu_2}{2} \left(\frac{\mu_1 - \mu_2}{\mu_1} \right) \times 2\mu_1$$

But $\frac{\mu_1 + \mu_2}{2}$ is nearly equal to μ_1

and $\frac{\mu_1 - \mu_2}{\mu_1} = \Delta$ (Relative refractive index difference)

$$\therefore \mu_1^2 - \mu_2^2 = 2\mu_1^2 \Delta$$

The quantity $\frac{\mu_1 - \mu_2}{\mu_1} = \Delta$ is known as relative refractive index difference or fractional refractive index change.

$$u = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

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∴ we have

$$N.A. = \sqrt{n_1^2 - n_2^2}$$

$$N.A. = n_1 \sin \theta$$

n_1 = core refractive index

Normalized Frequency (V-number)

The V-number or normalized frequency of an optical fibre is given by relation

$$V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

where a = radius of core

λ = wave length of light

and

$$\text{No. of mode of propagation} = \frac{V^2}{2}$$

Mode Of Propagation

A mode of propagation is a path that a light wave can follow as it travels down the core of optical fibre. There are two types of mode of propagation in a optical fibre

i) Single Mode Fibre

In a single mode optical fibre it consists of a very thin core of uniform refractive index surrounded by cladding of lower refractive index. The refractive index suddenly changes at core cladding interface in a single step so this is the single mode fibre. The core diameter is about $4-5 \mu\text{m}$ (micrometer). In this mode light travels along a axis.

ii) Multimode Step Fibre

It is similar to single mode step fibre but it has a large core diameter of about $100 \mu\text{m}$. As the core diameter is large, light travels in a fibre along zigzag path.

Q. In an optical fibre core material have $n_1 = 1.6$ and refractive index of cladding material is 1.3 what is the value of critical angle and calculate the angle of acceptance of cone.

→ Solution:

$$n_1(\text{core}) = 1.6$$

$$n_2(\text{cladding}) = 1.3$$

$$\sin c = \frac{n_2}{n_1}$$

$$c = \sin^{-1} \left(\frac{1.3}{1.6} \right) = 54.34^\circ$$

$$\sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

$$\therefore \theta_a = 68.86^\circ$$

acceptance angle

$$= 2\theta_a$$

$$= 137.73^\circ$$

Q. A step index fibre has a core of diameter $29 \mu\text{m}$, refractive index 1.52 and fractional refractive index difference of 0.0007 . It is operated as a wave length $1.3 \mu\text{m}$. Find Normalized frequency of fibre and no. of mode of propagation in a fibre.

→ Solution

$$\text{diameter } (d) = 29 \mu\text{m}$$

$$\text{radius } (a) = \frac{d}{2} = 14.5 \mu\text{m}$$

$$A = \frac{n_1 - n_2}{n_1} = 0.0007$$

$$\text{Wave length } \lambda = 1.3 \mu\text{m}$$

$$n_1 = 1.52$$

We have,

$$0.0007 = \frac{1.52 - n_2}{1.52}$$

$$\therefore n_2 = 1.51 \mu\text{m}$$

Then,

$$V_{\text{number}} = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2}$$

$$= \frac{2\pi \times 14.5}{1.3} \sqrt{(1.52)^2 - (1.51)^2}$$

$$= 4$$

and

$$\text{Mode of propagation} = \frac{V^2}{2}$$

$$= \frac{4^2}{2}$$

$$= 8 \text{ modes}$$

Unit-1

Elasticity [6]



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→ The property of a material body to recover its original state or condition on removal of the external forces to which it may be subjected is called elasticity.

Stress

→ When a body is subjected to a deforming force a restoring force occurs in the body which is equal in magnitude but opposite in direction with a applied force. This restoring force per unit area is known as stress.

If 'F' be the force applied and 'A' is the area of cross-section of the body then stress is found to be

∴

$$\text{Stress} = \frac{\text{Force}}{\text{Area}}$$

Strain

→ Strain is defined as change in dimension of body measured per unit dimension.

Hooke's Law

→ It states that, "Stress is proportional to strain within the elastic limit"
i.e. $\text{Stress} \propto \text{Strain}$

$$\Rightarrow E = \frac{\text{Stress}}{\text{Strain}}$$

This constant 'E' is called coefficient of elasticity or modulus of elasticity. The different types of coefficient of elasticity are:

1) Young's Modulus

It is defined as the ratio of the longitudinal stress to longitudinal strain within elastic limit.

$$Y = \frac{\text{Longitudinal Stress (Tensile)}}{\text{Longitudinal Strain}}$$

If 'l' be the length of wire, e be the extension produced due to applied force F on area A of wire then,

$$\text{Longitudinal Stress} = \frac{F}{A}$$

$$\text{Longitudinal strain} = \frac{e}{l}$$

$$\therefore Y = \frac{F}{A}$$

$$\frac{e}{l}$$

$$\therefore Y = \frac{F \times l}{A \times e} \quad \#$$

2) Bulk Modulus

It is defined as the ratio of stress to volumetric strain.

ie.

$$\text{Bulk Modulus (K)} = \frac{\text{Stress}}{\text{Volumetric strain}}$$

$$K = - \frac{F}{A} \div \frac{\Delta V}{V}$$

$$\therefore K = - \frac{F \times V}{A \Delta V}$$

3) Modulus of Rigidity

It is defined as the ratio of the tangential stress to shearing strain.

$$\text{ie. Modulus of rigidity } (\eta) = \frac{\text{tangential stress}}{\text{shearing strain}}$$

$$\eta = \frac{F}{A} \div \theta$$

$$\therefore \eta = \frac{F}{A \times \theta}$$

Poisson's Ratio

The ratio between lateral strain and linear strain due to a given stress within elastic limit called Poisson's ratio.

ie.

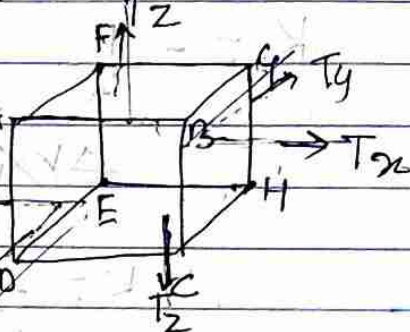
Poisson's ratio (σ) = $\frac{\text{Lateral Strain [L to force]}}{\text{linear Strain [l to force]}}$

⊕ Relation Connecting The Elastic Constant

Relation Between Y , K And σ

Young's bulk and poisson's

Let us consider a unit cube ABCDEFGH with forces T_x , T_y and T_z acting normally on its opposite outward on its opposite pairs of faces lying along the axis x , y and z respectively.



If ' α ' be the increase per unit length per unit stress along the direction of stress and ' β ' be the contraction per unit length per unit stress perpendicular to the direction of the stress. Then we have elongation produced in AB due to T_x is αT_x .

$$\alpha = \frac{\Delta l}{\text{stress}}$$

$$\alpha \times \text{Stress} = \frac{\Delta l}{\text{stress}} \times \text{stress}$$

$$\alpha \times T_x = \Delta l$$

Also, contraction produced on it due to T_y and T_z is equal to βT_y and βT_z respectively. So, that length of edge AB becomes

$$1 + \alpha T_x - \beta T_y - \beta T_z$$

Similarly,
length of edge B_y becomes
 $1 + \alpha T_y - \beta T_x - \beta T_z$

length of edge B_z becomes,
 $1 + \alpha T_z - \beta T_x - \beta T_y$

So, that the volume of the cube increases to

$$V = (1 + \alpha T_x - \beta T_y - \beta T_z)(1 + \alpha T_y - \beta T_x - \beta T_z)(1 + \alpha T_z - \beta T_x - \beta T_y)$$

Neglecting product and square of small quantities α and β we get if $T_x = T_y = T_z$ clearly,
* Volume of the cube becomes
 $V = 1 + (\alpha - 2\beta)(T_x + T_y + T_z)\beta T$
 $(V_2) = 1 + (\alpha - 2\beta)3T$

Hence, volume strain setup in the cube
 $= \frac{\Delta V}{V} = \frac{V_2 - V_1}{V_1} = \frac{1 + (\alpha - 2\beta)3T - 1}{1}$

$$= (\alpha - 2\beta)3T$$

$\therefore$ The bulk modulus for the material of the cube (K) = $\frac{\text{stress}}{\text{volume strain}} = \frac{T}{3T(\alpha - 2\beta)}$

$$\therefore K = \frac{1}{3(\alpha - 2\beta)} \quad \#$$

If instead of tension 'T' outwards a pressure 'P' be applied to compress the cube the decrease in its volume will similarly be $3P(\alpha - 2\beta)$

$\therefore$ Volume strain $= 3P(\alpha - 2\beta)$

Again,

$$\text{Bulk modulus (K)} = \frac{P}{3P(\alpha - 2\beta)}$$

$$= \frac{1}{3(\alpha - 2\beta)}$$

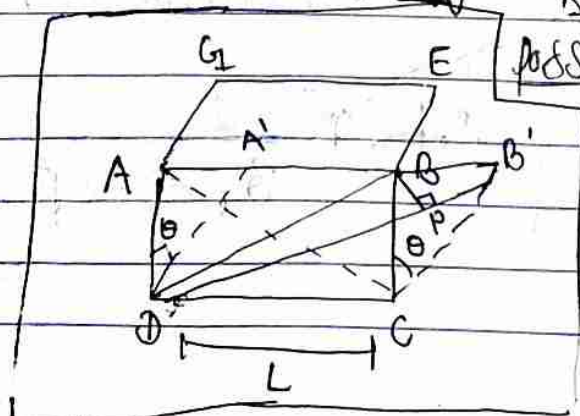
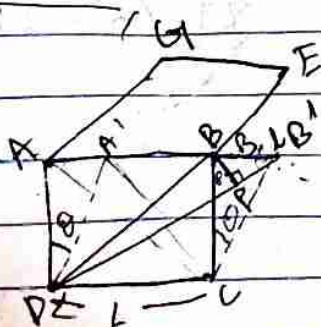
This may be written as;

$$K = \frac{1}{3(1 - 2\frac{\beta}{\alpha})}$$

Where $\gamma = \frac{1}{\alpha}$ and $\sigma = \frac{\beta}{\alpha}$

$$\therefore K = \frac{\gamma}{3(1 - 2\sigma)}$$

Relation Between γ , η & σ (Young's modulus of Rigidity & Poisson's Ratio.)



Let us consider upper face $ABEC$ of a cube of each edge of length ' L ' be sheared through an angle ' θ '. Under a tangential force F applied to it as shown in figure such that displacement $AA' = BB' = x$ with diagonal DB increased to DB' and diagonal AC decreased to $A'C$. Here, Shearing Stress $= \frac{F}{\text{Area } ABEC} = T$

Now,

A Shearing stress along $ABEC$ equivalent to tensile stress along DB and an equal compression stress CA at right $\angle$ ed.

If α and β are the longitudinal and lateral strain per unit stress respectively then,

Now,

Extension along diagonal DB due to tensile stress $= DB \times T \times \alpha$

and extension along diagonal DB due to compression stress along $AC = DB \times T \times \beta$

$$\begin{aligned} \therefore \text{Total extension along } DB &= DB \times T \times \alpha + DB \times T \times \beta \\ &= DB \times T (\alpha + \beta) \\ &= PB \text{ (represent)} \\ &= \frac{1}{2} L \times T (\alpha + \beta) \end{aligned}$$

Draw $BP \perp PB'$ then increase in length of diagonal DB can be taken approximately as equal to PB .

Since $\angle P$ is very small so $\angle AB'C = 90^\circ$ and $\angle BB'P = 45^\circ$

$$\begin{aligned} \therefore PB' &= BB' \cos 45^\circ \\ &= BB' \cos 45^\circ \\ &= BB' \times \frac{1}{\sqrt{2}} \\ &= \frac{l}{\sqrt{2}} \quad \text{--- (2)} \end{aligned}$$

From (1) and (2)

$$\sqrt{2} L \times T (\alpha + \beta) = l$$

$$\sqrt{2} \times \sqrt{2} L \times T (\alpha + \beta) = l$$

$$2 L \times T (\alpha + \beta) = l$$

$$T L (\alpha + \beta) = \frac{l}{2}$$

$$\frac{T \cdot l}{l} = \frac{1}{2(\alpha + \beta)}$$

$$\text{or } \frac{T}{\frac{1}{L}} = \frac{1}{2(\alpha + \beta)}$$

$$\text{or } \frac{T}{\theta} = \frac{1}{2\alpha(1 + \beta/L)}$$

$$\text{Here } \eta = \frac{T}{\theta}, \quad \frac{1}{\alpha} = \gamma, \quad \frac{\beta}{\alpha} = \sigma$$

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$$\eta = \frac{Y}{2(1+\sigma)}$$

Relation Between K , η And σ .

From above relations:

$$K = \frac{Y}{3(1-2\sigma)}$$

$$Y = 3K(1-2\sigma) \text{ --- (1)}$$

Also,

$$\eta = \frac{Y}{2(1+\sigma)}$$

$$2\eta(1+\sigma) = Y \text{ --- (2)}$$

Equating (1) and (2)

$$3K(1-2\sigma) = 2\eta(1+\sigma)$$

$$3K - 6\sigma = 2\eta + 2\eta\sigma$$

$$3K - 6\sigma - 2\eta - 2\eta\sigma = 0$$

$$3(K - 2\sigma) = 2\eta(1+\sigma)$$

Similarly A also require principle #



Limiting Value Of Poisson's Ratio σ ;

We have $3K(1-2\sigma) = 2\eta(1+\sigma)$

Since bulk modulus (K) and coefficient of rigidity (η) are both +ve quantities. It follows therefore, that

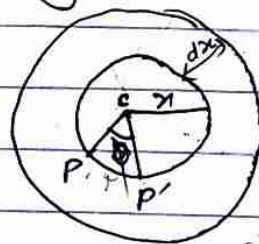
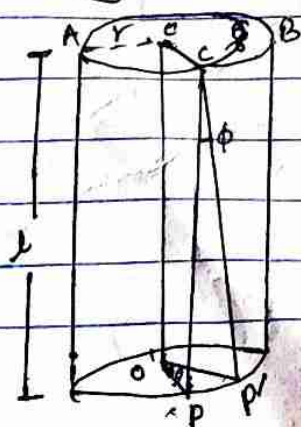
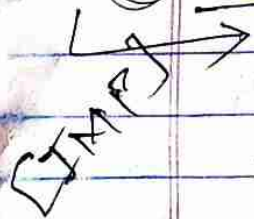
① If σ be +ve quantity the expression on the right hand side in the relation above will be +ve. The expression on left hand side too must be +ve. This is obviously possible when $2\sigma < 1$

$$\Rightarrow \sigma < \frac{1}{2}$$

$$\Rightarrow \sigma < 0.5$$

② If σ be -ve quantity the expression on left hand side above relation will be +ve. Hence, the expression right hand side too must be +ve. This is obviously possible when σ be not less than -1. Thus theoretically limiting value of σ are -1 and 0.5 though in actual practice it lies between 0.2 and 0.4 for most of the materials.

Twisting Couple On A Cylinder;



$$\theta = \frac{\phi}{L}$$

$$r = \frac{D}{2}$$

Consider a cylinder of radius 'r' and length 'l' clamped at upper end and free at lower end. If the twisting couple is applied on lower end then each cross-section of a cylinder twist through an angle ' θ '. Suppose cylinder consists

Suppose cylinder consists of large number of cylindrical shells. Consider one such shell of radius x and thickness dx . As the cylinder gets twisted the line CP takes new position CP' such that angle PCP' = ϕ . From figure we get,

$$\phi = \frac{PP'}{CP}$$

$$\text{or, } \phi = \frac{PP'}{x} \quad (PP' = r\theta)$$

$$\phi = \frac{r\theta}{x} \quad \text{--- (1) (For shell } PP' = r\theta)$$

This is shear strain produced in the shell

If 'F' be the tangential force applied at lower end of shell then, tangential stress is given by,

$$\text{Tangential Stress} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Tangential Stress} = \frac{F}{2\pi x dx} \quad \text{--- (2)}$$

If ' η ' is modulus of rigidity then

$$\eta = \frac{\text{tangential stress}}{\text{shear strain}}$$

$$= \frac{F \times l}{2\pi r \times dr \times 2\pi r \times \theta}$$

$$r \times 2\pi r \times dr \times \theta = F$$

$$\therefore F = \frac{2\pi r^2 dr \theta}{l} \quad (3)$$

This is force act on lower end of the shell.

Now, moment of this force about axis of shell is given by.

$$T = \frac{2\pi r^2 dr \theta \times r \times l}{l}$$

$$\int T = \frac{2\pi \eta \theta}{l} \int r^3 dr \quad (4)$$

This is twisting couple act on cylindrical shell of radius 'r' and thickness 'dr'. The twisting couple act on whole cylinder equal to

$$T = \int_0^R \frac{2\pi \eta \theta}{l} r^3 dr$$

$$= \frac{2\pi \eta \theta}{l} \int_0^R r^3 dr$$

$$= \frac{2\pi \eta \theta}{l} \times \frac{r^4}{4}$$

~~$$= \frac{2\pi \eta \theta}{l} \times \frac{r^4}{4}$$~~

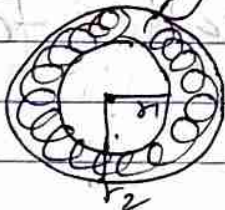
$$\therefore T = \frac{1}{2} \pi \eta \theta r^4 - (5)$$

The twisting couple per unit twist of wire or cylinder also called torsional rigidity of the material is given by

$$\text{Torsional rigidity (C)} = \frac{1}{2} \times \pi \eta \theta r^4$$

$$\therefore C = \frac{\pi \eta \theta r^4}{2l} - (6)$$

For hollow cylinder,



Suppose a hollow cylinder of internal radius r_1 and external radius r_2 then,

$$T = \frac{2\pi \eta \theta}{l} \int_{r_1}^{r_2} r^3 dr$$

Twisting couple on whole cylinder equal to

$$T = \int_{r_1}^{r_2} \frac{2\pi \eta \theta}{l} r^3 dr$$

$$T = \frac{2\pi\eta\theta}{l} \int_{r_1}^{r_2} r^3 dr$$

$$T = \frac{2\pi\eta\theta}{l} \left[\frac{r^4}{4} \right]_{r_1}^{r_2}$$

$$T = \frac{2\pi\eta\theta}{l} \left[\frac{r_2^4}{4} - \frac{r_1^4}{4} \right]$$

$$T = \frac{2\pi\eta\theta r_2^4}{4l} - \frac{2\pi\eta\theta r_1^4}{4l}$$

$$T = \frac{1}{2} \left(\pi \right)$$

$$\therefore T = \frac{1}{2} \pi \eta \theta (r_2^4 - r_1^4)$$

The twisting couple per unit twist of wire or cylinder also called torsional rigidity of the material

$$C = \frac{1}{2} \pi \eta \theta (r_2^4 - r_1^4)$$

$$\therefore C = \frac{1}{2} \pi \eta (r_2^4 - r_1^4) \quad \text{--- (7)}$$

Above relation is twisting couple per unit twist.

Q.1 Explain why a hollow cylinder is stronger than a solid cylinder of the same length, mass and material (missa)

Q.2 Two cylinder have the same length and mass and are made of same material; one is solid while other which is hollow has an external radius twice the internal radius. Compare the torsional rigidity of two cylinders.

Q.3 Find the greatest length of a steel wire that can hang vertically without breaking.
 Breaking stress for steel = 7.9×10^9 dynes/cm²
 Density of steel = 7.9 g/cm³

Ans

Q.1 Hollow cylinder is stronger than a solid cylinder of same length, mass and material. One is solid and another is hollow or tube like cylinder of inner radius r_1 and outer radius r_2 So,

Torsional couple for solid cylinder per unit twist

$$C_s = \frac{\pi \eta r^4}{2l}$$

Torsional couple for hollow cylinder per unit twist $C_h = \frac{\pi \eta (r_2^4 - r_1^4)}{2l}$

$$\therefore \frac{C_h}{C_s} = \frac{r_2^4 - r_1^4}{r^4} = \frac{(r_2^2 - r_1^2)(r_2^2 + r_1^2)}{r^4}$$

$$r_2^2 + r_1^2 = 2r^2 + 2r_1^2$$

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$$\Rightarrow r_2^2 + r_1^2 > r^2$$

The cylinder are of same length and mass,

$$\pi(r_2^2 - r_1^2) \rho l = \pi r^2 \rho l$$

$$r_2^2 - r_1^2 = r^2$$

So,

$$\frac{C_h}{C_s} = \frac{(r_2^2 - r_1^2)(r_2^2 + r_1^2)}{r^4}$$

$$\therefore \frac{C_h}{C_s} = \frac{r_2^2 + r_1^2}{r^2} - 1 \quad \left[\begin{array}{l} \text{Here, } r_2^2 - r_1^2 = r^2 \\ \Rightarrow r_2^2 = r^2 + r_1^2 \end{array} \right]$$

Thus, $C_h > C_s$ so, torsional rigidity for hollow cylinder is greater than for the solid cylinder of same mass, length and material.

Q. No 2 If r is

Solution:

If r is the radius of solid cylinder, r_2 and r_1 be outer and inner radius of hollow cylinder then,

$$\pi r^2 \rho l = \pi(r_2^2 - r_1^2) \rho l$$

$$r^2 = r_2^2 - r_1^2$$

But according to given that

$$r_2 = 2r_1$$

So,

$$\pi r^2 \rho l = \pi((2r_1)^2 - r_1^2) \rho l$$

$$\pi r^2 \rho l = \pi 3r_1^2 \rho l$$

$$\therefore r^2 = 3r_1^2$$

Torsional rigidity for ^{solid} hollow cylinder

$$C_s = \frac{\pi \eta r^4}{2l} = \frac{\pi \eta}{2l} \times 3r_1^2 \times 3r_1^2$$

$$C_s = \frac{\pi \eta}{2l} \times 9r_1^4 \quad \text{--- (1)}$$

Torsional rigidity of hollow cylinder

$$C_h = \frac{\pi \eta}{2l} [r_2^4 - r_1^4]$$

$$= \frac{\pi \eta}{2l} [(2r_1)^4 - r_1^4]$$

$$= \frac{\pi \eta}{2l} \times 15r_1^4 \quad \text{--- (2)}$$

Dividing (2) by (1)

$$\frac{C_h}{C_s} = \frac{\frac{\pi \eta}{2l} \times 15r_1^4}{\frac{\pi \eta}{2l} \times 9r_1^4}$$

$$\boxed{\frac{C_h}{C_s} = \frac{5}{3}}$$

Hollow cylinder has a great torsional rigidity than solid cylinder of same mass, length and material

Q.No 3) Solution:

Weight of steel wire (mg) = $A \times l \times \rho \times g$

Breaking Stress = $7.9 \times 10^9 \text{ dynes/cm}^2$

Density of steel (ρ) = 7.9 gm/cc

Now,

$$\text{Breaking Stress} = \frac{W}{A}$$

$$7.9 \times 10^9 = \frac{A \times l \times \rho \times g}{A}$$

$$\frac{7.9 \times 10^9}{7.9 \times 980} = l$$

$$\therefore l = 102040.8163 \text{ cm}$$

$$\text{Torsional rigidity (C)} = \frac{\pi \eta r^4}{2l}$$

$$Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\Delta l/l} = \frac{F \times l}{A \times \Delta l} = \frac{FL}{A \Delta l}$$

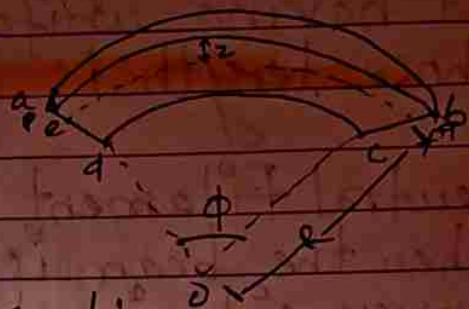
$$\text{Bulk modulus (K)} = \frac{\text{Stress}}{\text{Volumetric strain}}$$

$$\text{Volumetric strain}$$

Bending Of Beam

- > **Beam**: It is a cylindrical or rectangular bar or rod having uniform cross-sectional area whose length is much more than its thickness.
- > **Neutral Filament And Neutral Surface**
When the beam is bent the filament on a convex side is elongated and filament on a concave side is shortened. The filament line on the middle plane remains unchanged is called Neutral filament. The plane containing neutral filament is called neutral surface.
- > **Plane Of Bending**
The plane in which bending takes place is called plane of bending. The plane of bending is vertical if the beam is placed horizontally.
- > **Neutral Axis**
The line of intersection of neutral surface and plane of bending is called neutral axis.
- > **Bending Moment**
When external moment is applied on a beam then the internal moment is induced in the beam due to its elastic properties this moment called restoring moment. In equilibrium restoring moment is equal and opposite to external moment. The magnitude of resto

ring moment induced in beam called bending moment.



Let $abcd$ be a small section of beam and ef is a neutral filament and pq is a filament at a distance z from ef . If R is the radius of curvature for ef and ϕ is the angle subtended by ef at centre 'O'. Then from figure,

$$ef = R\phi \text{ and}$$

$$pq = (R+z)\phi$$

Now, elongation in filament pq is given by

$$\Delta l = pq - ef$$

$$\Delta l = R\phi + z\phi - R\phi$$

$$\therefore \Delta l = z\phi$$

Now, longitudinal strain produced is given by

$$= \frac{\text{change in length}}{\text{original length}}$$

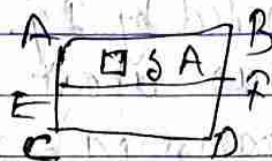
$$= \frac{z\phi}{R\phi}$$

$$\text{Strain} = \frac{z}{R}$$

If Y is the Young's modulus of beam then

$$Y = \frac{\text{Stress}}{\text{Strain}}$$

$$\begin{aligned} \text{Stress} &= Y \times \text{strain} \\ &= Y \times \frac{z}{R} \end{aligned}$$



Consider a section ABCD of the strip at right angle to its length and the plane of bending. Then the force acting on the strip are perp. to this section and the line EF lies on the neutral surface. The forces producing elongation acting in the upper half ABEF and those producing contraction act in the lower half CDEF in opposite direction perp. to the section ABCD and hence constitute a couple.

To find the moment of this couple consider a small area ΔA lying at a distance z from the neutral axis EF. Then force on area,

$$\begin{aligned} \Delta F &= \text{Stress} \times \Delta A \\ &= \frac{Yz}{R} \times \Delta A \end{aligned}$$

∴ Moment of the force about the axis EF =

$$\begin{aligned} &= \frac{Yz}{R} \times \Delta A \times z \\ &= \frac{Yz^2 \Delta A}{R} \end{aligned}$$

Hence, moment of all the surface acting at various point of the whole face ABCD = $\frac{y}{R} \sum s a z^2$

The quantity $\sum s a z^2$ is called geometrical moment of inertia of cross section of the beam which is denoted by I_g . Therefore, total moment of all the forces about EF or the bending moment of the beam, $M = \frac{y I_g}{R}$

For a beam of rectangular cross section,

$$I_g = \frac{b d^3}{12}$$

where, b = breadth and d = depth of beam

Then, bending moment is

$$M = \frac{y b d^3}{12 R}$$

And for a beam of circular cross section, $I_g = \frac{\pi r^4}{4}$

where, r = radius of circular cross section,

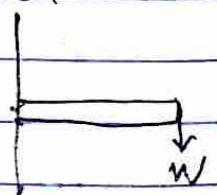
$\therefore$ Bending moment is

$$M = \frac{y \pi r^4}{4 R}$$

Cantilever

It is a beam fixed horizontally at one end and loaded at the other

$$\frac{5y}{8} = \frac{1}{8} \left[\frac{w}{1 + \left(\frac{w}{2} \right)^2} \right]$$

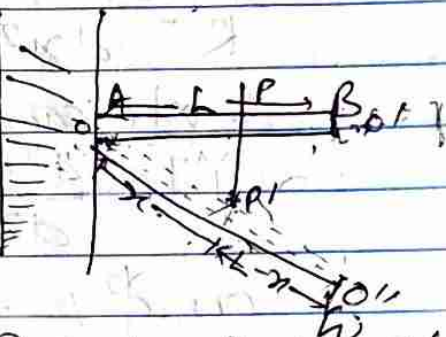


$\frac{d^2y}{dx^2}$
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Depression of Cantilever Loaded At Free End

Let AB be a cantilever of length L rigidly fixed at end A and loaded with a weight ' w ' at its free end so the neutral axis oo' will take up the new position oo'' so that the def. depression of loaded end is $o'o''$



Now, consider a beam at P which is at a distance x from O . Then the moment of couple due to load will be given by

Restoring couple

$$w \times plo'' = wx(L-x)$$

This, as we know is in equilibrium with the bending

i.e. balanced by the bending moment $\frac{Y I_g}{R}$ where,

Y = Young's modulus for the material of beam

I_g = Geometrical moment of inertia of its cross-section and

R = Radius of curvature of the neutral axis of the section at P .

Therefore, we have

$$\text{Deforming couple} = \text{Restoring couple} \left(\frac{Y I_g}{R} \right)$$

$$\frac{Y I_g}{R} = wx(L-x) \quad (1)$$

Now, the curvature $\frac{1}{R} = \frac{d^2y}{dx^2}$

Here, $\frac{dy}{dx}$ is small quantity. Hence, $\left(\frac{dy}{dx} \right)^2$ is neglected.

Therefore,

$$\frac{1}{R} = \frac{d^2 y}{dx^2}$$

$\therefore$ We can write Eqn (1) as

$$Y I_g \frac{d^2 y}{dx^2} = \omega (L - x)$$

or $\frac{d^2 y}{dx^2} = \frac{\omega (L - x)}{Y I_g}$

Integrating both side we get

$$\frac{dy}{dx} = \frac{\omega}{Y I_g} \left[Lx - \frac{x^2}{2} \right] + C_1$$

Where C_1 = integration constant

Since end A is fixed we have

At $x=0$

$$\frac{dy}{dx} = \frac{\omega}{Y I_g} \left[L \times 0 - \frac{0^2}{2} \right] + C_1$$

$$\therefore \frac{dy}{dx} = 0$$

$$\therefore C_1 = 0$$

Hence, above equation becomes

$$\frac{dy}{dx} = \frac{\omega}{Y I_g} \left[Lx - \frac{x^2}{2} \right]$$

Again, integrating we get

$$y = \frac{\omega}{Y I_g} \left[\frac{Lx^2}{2} - \frac{x^3}{6} \right] + C_2$$

Where C_2 = another integration constant

$$y=0 \therefore C_2=0$$

Again at $x=0$, $y=0$

$$\therefore C_2 = 0$$

Hence, $y = \frac{\omega}{YI_g} \left[\frac{Lx^2}{2} - \frac{x^3}{6} \right]$

Where y gives depression of beam at a distance x from fixed end A.

Since, the free end B of the beam is at a distance L from A we have,

$$x = L$$

$\therefore$ depression of loaded end be

$$y = \frac{\omega}{YI_g} \left[\frac{L^3}{2} - \frac{L^3}{6} \right]$$

$$\therefore y = \frac{\omega L^3}{3YI_g}$$

Since for a beam of rectangular cross-section of breadth ' b ' and depth ' d '.

$$I_g = \frac{bd^3}{12}$$

$$\therefore y = \frac{\omega L^3}{3 \times \frac{bd^3}{12} \times Y}$$

$$\therefore y = \frac{4\omega L^3}{Ybd^3}$$

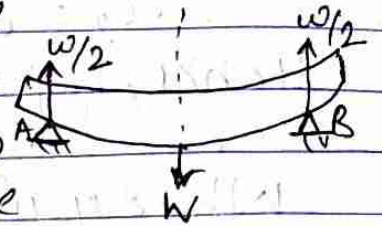
And for a beam of circular cross section of radius ' r '.

$$I_g = \frac{\pi r^4}{4}$$

$$\therefore y = \frac{\omega L^3}{3Y \times \frac{\pi r^4}{4}}$$

$$\therefore y = \frac{4\omega L^3}{3\pi r^4 Y}$$

Bar Supported At Two Ends And Loaded At The Middle



Let us consider a rod which is supported at two knife edges A and B and loaded at the centre with a load w . So that the reaction at knife edges be $w/2$ in upward direction.

Since the middle of the rod may be considered practically horizontal we may consider the above arrangement as equal to inverted cantilever fixed at centre and loaded at A and B with a load $w/2$ in upward direction. If ' l ' be the length of the beam or rod. Then the depression y is given by

$$y = \frac{w/2 \times (l/2)^3}{3 \times Y \times I_g}$$

$$\therefore y = \frac{wl^3}{48YI_g}$$

OK

2075) # Compare the load required to produce equal depression for two beams made of same material, same length and weight with only difference that one has circular cross section while the cross-section of other is square

Ans // Consider a bar of each length 'L', ρ be its density, r is the radius of curvature and a be the side of the square.

Show,

$$\text{Mass of square} = \rho L a^2$$

$$\text{Mass of circular} = \rho \pi r^2 L$$

$\therefore$ Both have equal weight then,

$$\rho L a^2 = \rho \pi r^2 L$$

$$\text{Geometric } \pi r^2 = a^2$$

If I_1 is the M.I of square bar then,

$$\text{Depression for square bar } y = \frac{W_1 L^3}{3 \pi I_1} \quad (1)$$

and

$$I_2 \text{ is the } \text{Geometric } \text{M.I of circular bar } y = \frac{W_2 \times L^3}{3 \pi I_2} \quad (2)$$

From (1) and (2)

$$\frac{W_1 \times L^3}{3 \pi I_1} = \frac{W_2 \times L^3}{3 \pi I_2}$$

$$\frac{W_1}{I_1} = \frac{W_2}{I_2}$$

$$\therefore I_1 = \frac{a^4}{12} \text{ and } I_2 = \frac{\pi r^4}{4}$$

$$\therefore \frac{W_1}{W_2} = \frac{I_1}{I_2}$$

$$\frac{W_1}{W_2} = \frac{a^4}{12} \times \frac{4}{\pi r^4}$$

$$= \frac{4a^4}{12\pi r^4}$$

$$= \frac{a^4}{3\pi r^4}$$

$$= \frac{a^4}{3\pi r^4}$$

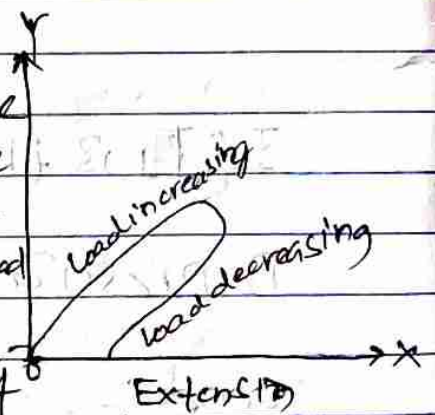
$$= \frac{\pi^2 r^4}{3\pi r^4} \quad (\because a^2 = \pi r^2)$$

$$= \frac{\pi}{3}$$

$$\therefore \frac{W_1}{W_2} = \frac{\pi}{3} = 1.05$$

Elastic Hysteresis

Q. This is purely a consequence of elastic after effect due to which the strain persists even when the stress is removed and thus lags behind the stress. As a result if a material be subjected to a full cycle of increasing and decreasing load the strain is found to be greater for the same value of the stress when it is being unloaded than it is being loaded as shown in figure. The lagging behind of strain called elastic hysteresis.



Unit-2

Surface Tension [9]

Q.75) It is defined as the force per unit length acting on either side of a line drawn in the liquid surface. In equilibrium the direction of the force being tangential to the surface and perpendicular to the line.

Surface Energy

The surface of a liquid is in state of a tension the force of a surface tension tends to decrease the surface area. Hence when the area of a liquid surface is increased work is done against surface tension. This work is stored in the surface as surface energy.

Consider a wire frame ABCD in which the wire AB is movable form a soap film over it. The wire AB is pulled inward due to surface tension by a force $T \times 2l$ where T = surface tension and l is the length AB. The factor 2 appears

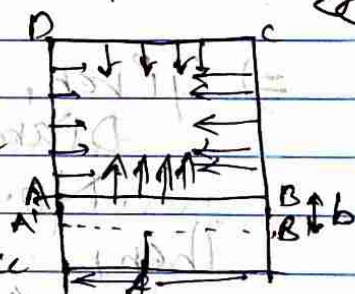


Fig. 1

because there are two surfaces of the soap film. If the film is pulled by a small distance b to the position $A'B'$ keeping the temperature constant then the work done is given by $= T \times 2l \times b$

Increase in area $= 2l \times b$

Surface energy = $\frac{\text{work done in increasing surface area}}{\text{Increase in surface area}}$

$$= T \times 2L \times b$$

$$2L \times b$$

$$= T$$

∴ Surface energy of T

This energy is stored in the surface. Hence, surface energy per unit area of a surface is numerically equal to the surface tension.

What amount of energy will be liberated if 1000 droplets of water each of 10^{-8}m in diameter collapse to form a one large spherical drop. Surface tension of water $75 \times 10^{-3} \text{ N/m}$.

⇒ Given,

Diameter of small droplet (d) = 10^{-8}m

No. of drops (n) = 1000

Then,

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3$$

$$\frac{4}{3} \times \pi \times R^3 = 1000 \times \frac{4}{3} \times \pi \times \left(\frac{d}{2}\right)^3$$

$$R^3 = 1000 \times \left(\frac{d}{2}\right)^3$$

$$R^3 = 1000 \times \left(\frac{1 \times 10^{-8}}{2}\right)^3$$

$$\therefore R = 5 \times 10^{-8}\text{m}$$

$$\begin{aligned}
 \text{Area of } \overset{1000}{\text{small drops}} (A_1) &= 4\pi r^2 \times 1000 \\
 &= 4 \times \pi \times \left(\frac{d}{2}\right)^2 \times 1000 \\
 &= 4 \times \pi \times \left(\frac{1 \times 10^{-8}}{2}\right)^2 \times 1000 \\
 &= 3.14 \times 10^{-16} \text{ m}^2 \times 1000 \\
 &= 3.14 \times 10^{-13} \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of large drops } (A_2) &= 4\pi R^2 \\
 &= 4 \times \pi \times (5 \times 10^{-8})^2 \\
 &= 3.14 \times 10^{-14} \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Difference in surface area } (\Delta A) &= 3.14 \times 10^{-13} - 3.14 \times 10^{-14} \\
 &= 2.826 \times 10^{-13} \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Surface energy} &= \Delta A \times \gamma \\
 &= 2.826 \times 10^{-13} \text{ J} \left[\frac{\text{N}}{\text{m}} \right]
 \end{aligned}$$

Excess Pressure Inside A Drop

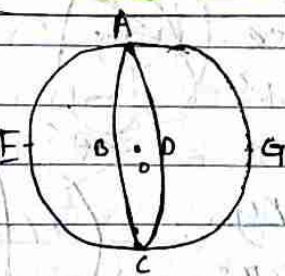


Fig: 1

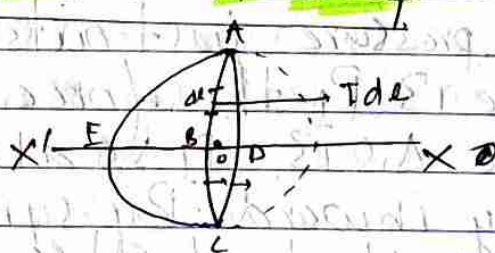


Fig: 2

Let us consider a spherical drop of liquid radius 'R'. Imagine a diametric cross-section ABCD of the drop which divides the drop into two hemisphere. The surfaces of the two hemisphere touch each other along the perimeter ABCD. Consider a equilibrium of the hemispherical surface ABCDE. Forces acting on this surface are

① F_1 due to surface tension of the surface ABCDE in contact

② F_2 due to the air outside the surface ABCDE.

③ F_3 due to the liquid inside the surface ABCDE.

The force on any small part ~~of~~ of the periphery is $T \delta s$ and acts parallel to symmetry axis OX. The resultant of all these force due to surface tension is

$$F_1 = 2\pi CRT \text{ along OX axis}$$

Now, consider the force due to air outside the surface ABCDE. Consider a small part δs of the surface as shown in fig-3

If the pressure just outside the surface is P_1 the force on this surface δs is $P_1 \delta s$ along the radius inward. By symmetry the resultant of all such surface forces act along OX. If the radius through δs makes an

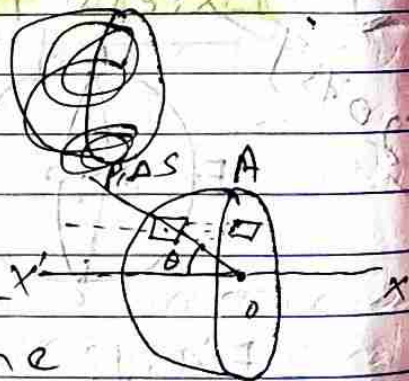


Fig 3:

angle θ with OX the component of $P_1 \delta s$ along OX will be

$P_1 \delta s \cos \theta$. If we project area δs on the diametric plane ABCD the area of the projection will be $\delta s \cos \theta$. Thus we can write component of $P_1 \delta s$ along OX

$$= P_1 \times (\text{projection of } \delta s \text{ on diametric plane } ABCD)$$

$$= P_1 \times$$

The resultant force due to the air outside the hemispherical surface is

$$F_2 = P_1 \times (\text{Projection of hemi-spherical surface ABCDE on the plane ABCD})$$

$$\Rightarrow F_2 = P_1 \times \pi R^2 \text{ along OX}$$

Similarly The resultant force on this surface due to liquid inside the surface is

$$F_3 = P_2 \times \pi R^2 \text{ along OX}$$

Where P_2 is the pressure just inside the surface. This force will be in OX direction.

For equilibrium of the hemi-spherical surface ABCDE we should have

$$F_3 = F_1 + F_2$$

$$P_2 \times \pi R^2 = 2\pi R T + P_1 \times \pi R^2$$

$$P_2 \times \pi R^2 = \pi R (2T + P_1 R)$$

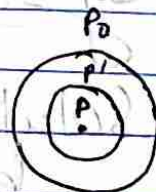
$$(P_2 - P_1) = \frac{2T}{R}$$

i.e. Pressure inside surface is greater than the pressure outside the surface by an amount

$$\frac{2T}{R}$$

Excess Pressure In A Soap Bubble

Let the pressure of air outside the bubble be P_0 that within the soap bubble is P_1 and that of the air inside the bubble be P .



Looking at outer surface the solution is on the concave side of the surface. Hence

$$\therefore P_0 - P = \frac{2T}{R} \quad \text{--- (1)}$$

Where R = Radius of the bubble.
Similarly, looking at the inner surface air is on the concave side of surface.
Hence $P - P_0 = \frac{2T}{R} \quad \text{--- (2)}$

Adding eqn (1) and (2)

$$P - P_0 = \frac{4T}{R} \quad \text{--- (3)}$$

#1 A 0.02 cm liquid column balance the excess pressure inside a soap bubble of radius 7.5 mm. Determine the density of the liquid.
Surface tension of soap solution = 0.03 N/m

#2 There is an air bubble of radius 1 mm in a liquid of surface tension 0.075 N/m and density 1000 kg/m³. The bubble is at depth of 10 cm below the free surface. By what amount is the pressure inside the bubble greater than the atmospheric pressure?
 $g = 9.8 \text{ m/sec}^2$

#1 Solution:

length of liquid column (h) = 0.02 cm = 0.0002 m
Radius of soap bubble (R) = 7.5 mm = 7.5 × 10⁻³ m

Surface tension of soap (T) = 0.03 N/m

Now,

$$\text{Excess pressure} = \frac{4T}{R} = 16 \text{ Pa}$$

Again,

$$P = \rho gh$$

$$16 = \rho \times 9.81 \times 0.0002$$

$$\rho = 8154.94 \text{ kg/m}^3$$

#2.

Radius of air bubble (R) = 1 mm
 $= 1 \times 10^{-3} \text{ m}$

Surface tension (T) = 0.075 N/m

Density (ρ) = 1000 kg/m^3

$$P = P_A + P_L$$

$$\therefore P_i - P = \frac{2T}{R}$$

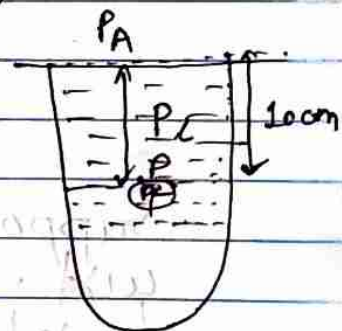
$$P_i - P_A - P_L = \frac{2T}{R}$$

$$P_i - P_A = \frac{2T}{R} + \rho gh$$

$$= \frac{2 \times 0.075}{1 \times 10^{-3}} + 1000 \times 9.81 \times 0.1$$

$$= 150 + 980$$

$$= 1130 \text{ Pa}$$



Angle of Contact

The angle between the tangent to the liquid surface at the point of contact and the solid surface inside the liquid is known as angle of contact.

Rise Of Liquid In Capillary Tube

[20/4]

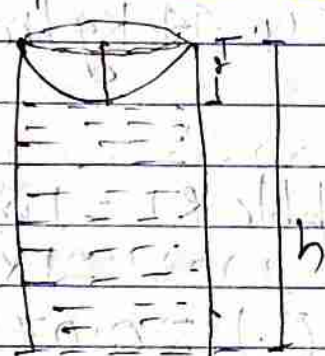


Fig 2

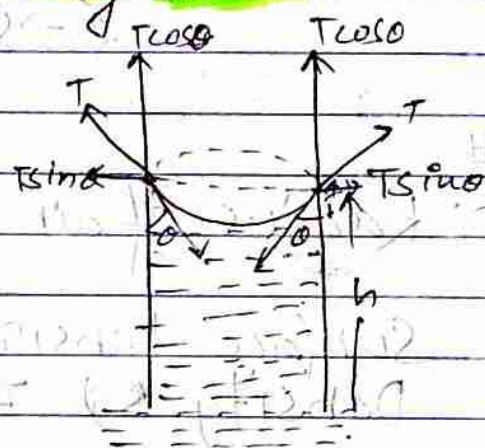


Fig 1

Suppose a capillary tube is dipped in a liquid. Let θ be angle of contact, h be the height to which the liquid rises, r be the internal radius of capillary tube, ρ be the density of the liquid and T be the surface tension.

Consider a small part dl of the periphery $2\pi r$ along which the surface of the liquid at which the tube meet. The liquid surface across this pulls the tube surface by a force $T dl$ tangentially along the liquid surface. From Newton's third law the tube surface across this small part

pull the liquid surface by an equal force T in opposite direction. The vertical component of this force is $T \cos \theta$. Therefore, the total force exerted on liquid surface by the cube surface across the contact circle is

$$F = \int T \cos \theta$$

$$\Rightarrow F = 2\pi r T \cos \theta$$

$$\therefore F = 2\pi r T \cos \theta$$

The horizontal component $T \sin \theta$ adds up to zero when summed over the entire periphery

Now, the volume of the liquid in the cube above the free surface of liquid is

$$\begin{aligned} V &= \text{Volume of liquid of height } h = \text{Vol. of hemisphere of radius } r \\ &= \pi r^2 h - \frac{1}{2} \times \frac{4}{3} \pi r^3 \quad (\because \text{hemisphere} = \frac{4}{3} \pi r^3 \times \frac{1}{2}) \\ &= \pi r^2 h - \frac{2}{3} \pi r^3 \\ &= \pi r^2 h - \frac{2}{3} \pi r^3 \end{aligned}$$

$$V = \pi r^2 (h - \frac{2}{3} r)$$

$$\therefore V = \pi r^2 (h - \frac{2}{3} r)$$

$$= \pi r^2 (h + r) - \frac{4}{3} \pi r^3$$

$$= \pi r^2 h + \pi r^3 - \frac{4}{3} \pi r^3$$

$$V = \pi r^2 (h + r - \frac{2}{3} r)$$

$$\therefore V = \pi r^2 (h + \frac{1}{3} r)$$

$$V = \pi r^2 (h + r - \frac{2}{3} r)$$

$$V = \pi r^2 h + \frac{1}{3} \pi r^3$$

$$V = \pi r^2 (h + \frac{1}{3} r)$$

$$\text{Now, Weight of liquid} = V \rho g \\ = \pi r^2 \left(h + \frac{r}{3} \right) \rho g$$

At Equilibrium

$$2\pi r T \cos \theta = \pi r^2 \left(h + \frac{r}{3} \right) \rho g$$

$$2T \cos \theta = r \left(h + \frac{r}{3} \right) \rho g$$

~~$$2T \cos \theta = r \left(h + \frac{r}{3} \right) \rho g$$~~

Neglecting $\frac{r}{3}$ then,

$$\boxed{\therefore h = \frac{2T \cos \theta}{\rho g}}$$

$$F = -\eta A \frac{dv}{dy}$$

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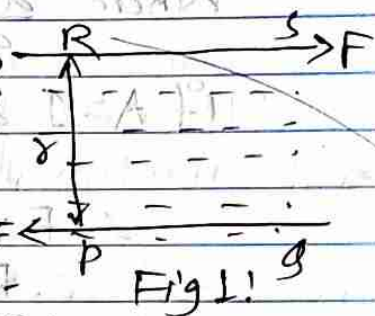
Unit-3

Viscosity [5]

The property of liquid by virtue of which a liquid opposes relative motion between different layer is called viscosity.

Coefficient of Viscosity

Let a layer RS of a liquid move with velocity 'v' relative to a parallel layer PQ which is at a distance 'r' from it. Let the force 'F' on an area 'A' required to produce this motion be 'F'. This force must act on RS in the direction of motion along velocity.



An equal force will, therefore, act on it in opposite direction due to viscosity. This backward dragging force 'F' depends upon following factors:

1) $F \propto -v$ (i.e. F is directly proportional to relative velocity 'v' and acts in a direction opposite to the direction of motion of the liquid)

2) $F \propto A$

3) $F \propto \frac{1}{r}$

Combining all factors we have

$$F \propto -\frac{vA}{r}$$

$$\therefore F = -\eta A \frac{v}{r}$$

where η = constant depending upon nature of liquid called coefficient of viscosity.

If two layers are very close together then denoting distance between them by dr and relative velocity dv then we have

$$F = -\eta A \frac{dv}{dr}$$

Where $\frac{dv}{dr}$ = velocity gradient

if $A=1$ and $dv/dr=1$ then,

$$\therefore F = -\eta$$

$$\therefore F = \eta \text{ (numerically)}$$

Thus coefficient of viscosity is defined as the tangential force per unit area required to maintain a unit velocity gradient i.e. a unit relative velocity between two layers at a unit distance apart

Stream Line Flow

If with time velocity at every point in the liquid remains constant in magnitude as well as dirⁿ then flow said to be steady. In other word in steady flow each particle follows exactly the same path and same velocity as as predecessor. In such a case, the liquid is said to have an orderly stream line flow. The line along which the liquid moves when the flow is steady known as streamline

$$\sqrt{\frac{r^3}{m^3}}$$

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Critical Velocity

✓ The critical velocity of liquid is that velocity of flow above which the flow ceases to be stream line.

Turbulent Flow

When the velocity of liquid is greater than the critical velocity. The flow of liquid doesn't remain steady but becomes turbulent. When flow is turbulent there are eddies and whirlpool in the motion path as well as velocities of particle are continuously changing.

Significance Of Reynold's Number

✓ The critical velocity of a liquid having coefficient of viscosity (η) and density (ρ) flowing through a tube of radius (r) is given by

$$V_c = \frac{Re \eta}{8r} \quad \text{--- (i)}$$

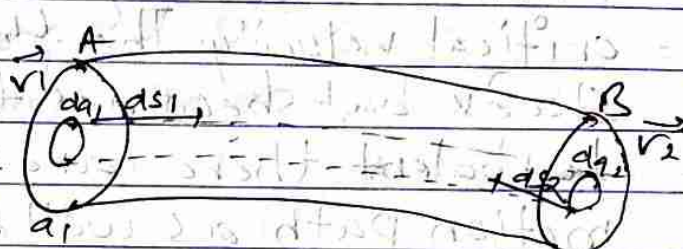
The significance of equation (i) lies in the fact that a non-dimensional quantity of the form $V_c r / \eta$ determines the process that takes place during the motion of the liquid through a tube and when this quantity attains a certain value the flow changes from a steady streamline flow to a turbulent one in which eddies are formed. Thus critical velocity of liquid is

- i) directly proportional to its viscosity
- ii) inversely " " " density
- iii) " " " radius of tube

Therefore, follows that narrow tubes, low density and high viscosity help in producing orderly motion and wide tube, high density and low viscosity tend to produce turbulent motion

Equation of Continuity

Consider a fluid to be flowing through a pipe AB with A_1 and A_2 as its cross-sectional areas at section A and B. Let $\vec{v}_1$ and $\vec{v}_2$ be the velocity at section A and B respectively.



Now, consider an infinitely narrow tube of flow of cross-sectional area dA_1 and dA_2 at section A and B respectively. Let the fluid covers distance ds_1 and ds_2 in time 'dt' at the two ends and

ρ_1 and ρ_2 be the densities of fluid at A and B

Now, mass of fluid entering the cube of flow per unit time at section A = $dA_1 \cdot ds_1 \cdot \rho_1$

$$= dA_1 \cdot \rho_1 \cdot v_1 \cdot dt$$

Mass of fluid leaving the cube of flow per unit time at section B

$$= dA_2 \cdot ds_2 \cdot \rho_2$$

$$= dA_2 \cdot \rho_2 \cdot v_2 \cdot dt$$

Mass of fluid entering the whole section A of the pipe per unit time

i.e. Mass-rate of fluid flow at A

$$\int_0^a da_1 \rho_1 v_1$$

$$= \rho_1 v_1 a_1$$

Similarly mass-rate of fluid flow at B

$$= \rho_2 v_2 a_2$$

For incompressible fluid $\rho_1 = \rho_2 = \rho$

From the law of conservation of matter,

Mass of fluid entering a section A per second = Mass of fluid leaving a section B per second (since there can be no source or sink)

$$\rho_1 v_1 a_1 = \rho_2 v_2 a_2 \quad (\because \rho_1 = \rho_2 = \rho)$$

$$\Rightarrow \therefore a_1 v_1 = a_2 v_2 \quad (\text{only for incompressible fluid})$$

(v = velocity)

This is called the equation of continuity. Thus equation of continuity states that, "the volume rate of flow of an incompressible fluid remains constant throughout the flow."

Poiseuille's Equation

- 1) The flow of liquid is parallel to the axis of the tube throughout and it is streamline.
- 2) The liquid flow is steady i.e. no accn of flow exist.
- 3) The velocity of the liquid layer in contact with the walls is zero and increase regularly and continuously towards the inner side, being max. along axis of tube.

Theory:

Let AB be a capillary tube of length 'l' and radius 'a' as shown in

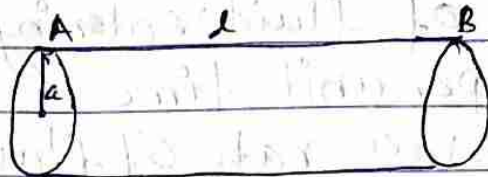


Fig: 1

fig: 1. Consider a cylindrical layer of the liquid coaxial with the tube of inner radius 'r' and outer radius $r+dr$ as shown in figure 2.



Fig: 2

Let velocity of the liquid at a distance 'r' from the axis of the tube is $\vec{v}$ and $\vec{v}+d\vec{v}$ is $v+dv$ therefore the velocity gradient is dv/dr .

The surface area of cylinder (2) = $2\pi r l$

The liquid on the inner side of this cylindrical layer is moving faster while that on the outer side is moving slower.

∴ The tangential force exerted by the outer layer on the inner layer opposite to the direction of motion is $(F) = -\eta (2\pi r l) \frac{dv}{dr}$

The forward push due to pressure difference Δp on two side of cylinder of radius 'r' = $\Delta p \pi r^2$

When the motion is steady there is no acceleration of liquid and thus we have,

$$\Delta p \pi r^2 = -\eta 2\pi r l \frac{dv}{dr}$$

$$\Delta p r = -\eta 2 l \frac{dv}{dr}$$

$$\Rightarrow dv = -\frac{pr}{2\eta l} dr$$

Integrating both side we have

$$\int_0^r dv = -\frac{pr}{2\eta l} \int_0^r r dr$$

$$v = -\frac{p}{2\eta l} \cdot \frac{r^2}{2} + C_1$$

C_1 = integrating constant

Here, as the velocity of liquid along the sides of the cube is zero i.e. at $r=a$ then $v \rightarrow 0$

$$0 = -\frac{p}{2\eta l} \cdot \frac{a^2}{2} + C_1$$

$$0 = -\frac{p}{2\eta l} \cdot \frac{a^2}{2} + C_1$$

$$\therefore \frac{p}{2\eta l} \cdot \frac{a^2}{2} = C_1$$

$\therefore$ Above eqn becomes

$$v = \frac{p}{4\eta l} [a^2 - r^2] \quad \text{--- (1)}$$

This gives velocity of flow at distance r from the axis of the tube to the cylindrical layer.

The area of cross-section of cylindrical layer of radius ' r ' and thickness dr is $= 2\pi r dr$

$\therefore$ Volume of liquid passing per second through this area $= 2\pi r dr \times v$

Hence, the volume of liquid passing through whole tube per second is given by $\int$

$$\int dr = \int_0^a 2\pi r dr \times \theta$$

$$\Rightarrow V = 2\pi \cdot \int_0^a r^2 dr \cdot \frac{P}{4\pi l} [a^2 - r^2]$$

$$V = 2\pi \times \frac{P}{4\pi l} \int_0^a r(a^2 - r^2) dr$$

$$V = 2\pi \times \frac{P}{4\pi l} \left[a^2 \int_0^a r dr - \int_0^a r^3 dr \right]$$

$$V = 2\pi \times \frac{P}{4\pi l} \times a^2 \times \frac{a^2}{2} - \frac{a^4}{4}$$

$$V = \frac{2\pi \times P}{24\pi l} \times \frac{a^4}{2} - \frac{a^4}{4}$$

$$\therefore V = \frac{\pi P a^4}{8\pi l}$$

$$F = 6\pi\eta r v$$

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Stoke's Law (Different medium) different molecule used spherical
If a spherical body of radius 'r' moving with the terminal velocity 'v' in a viscous medium of coefficient of viscosity ' η ' experience a viscous drag given by the relation

$$F = 6\pi\eta r v \text{ This is referred as Stokes law.}$$

Terminal Velocity

When a body falls through a liquid or gas it carries along with it the layer of fluid in contact and thus tends to produce a relative motion between the layers of fluid. The relative motion is opposed by forces due to viscosity. The opposing force increases with the velocity of the body and if the body is small it soon becomes equal to the driving force producing motion. The body then moves with a constant velocity known as terminal velocity.

Expression For Terminal Velocity

If 's' be the density of spherical body then it's

$$\text{weight} = \text{mass of body} \times g$$

$$= V \times s \times g$$

$$= \frac{4}{3} \pi r^3 s g$$

Where r = radius of spherical body

g = accn due to gravity

Since it displaces it's own volume of the medium then the weight of the displaced medium or the upthrust on it : $\frac{4}{3} \pi r^3 \sigma g$

$\therefore \sigma$ = density of the medium

∴ Resultant force on the body

$$i.e. \frac{4}{3} \pi r^3 (\rho - \sigma) g$$

When the body attains terminal velocity (i.e. $a=0$) this downward force must be balanced by the viscous drag

∴ we have

$$\frac{4}{3} \pi r^3 (\rho - \sigma) g = 6 \pi \eta r v$$

$$\frac{4}{3} r^2 (\rho - \sigma) g = 6 \eta v$$

$$\therefore v = \frac{2 r^2 (\rho - \sigma) g}{9 \eta}$$

Bernoulli's Theorem.

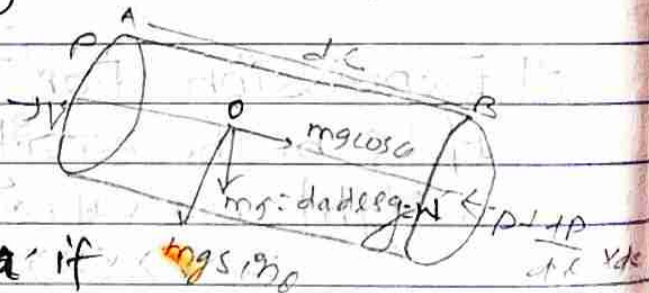
It states that "the total energy of an incompressible non-viscous fluid in steady flow remains constant throughout the flow".

Consider a infinitesimal small portion AB of a tube of flow of length 'dl' having uniform cross-section area 'da' if 'ρ' be density of fluid and its mass

$$m \text{ in portion} = \rho da dl$$

$$\therefore \text{Weight} = mg$$

$= \rho da dl g$ which acts vertically downward at its centre of gravity 'O' making an angle 'θ' with the direction of flow as shown in figure. The component $mg \cos \theta$ acts in the direction of flow and component $mg \sin \theta$ acts perpendicular to the



If 'P' be the pressure on face A of the tube then on face B the pressure will be $P + \frac{dP}{dx} \times dx$ in the shown direction where $\frac{dP}{dx}$ = pressure gradient

Therefore force acting on face A in forward direction = $P \cdot da$

Force acting on face B ^{in backward} = $(P + \frac{dP}{dx} dx) da$

Hence, the net force acting on mass $da \cdot dx$ of the fluid in the tube = $P da + mg \cos \theta - (P + \frac{dP}{dx} dx) da$

$$\Rightarrow F = - \frac{dP}{dx} \times dx da + mg \cos \theta \quad \text{--- (1)}$$

If 'v' be the velocity of the fluid as it enters the tube at A we have

acceleration = $\frac{dv}{dt}$ where dt = time taken by the fluid to cover the length dx of the cube

$\therefore$ Force acting on mass $da \cdot dx$ of the fluid in the tube is

$$F = \text{mass} \times \text{acceleration}$$

$$= da \cdot dx \times \frac{dv}{dt} \quad \text{--- (2)}$$

Now, taking the most general case where velocity being a function of both distance and time we have,

$$v(x, t)$$

$$dF = \frac{\partial F}{\partial x} \times dx + \frac{\partial F}{\partial t} \times dt$$

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial t} dt$$

$$\Rightarrow \frac{dv}{dt} = \frac{\partial v}{\partial x} \times \frac{dx}{dt} + \frac{\partial v}{\partial t}$$

$$= \frac{\partial v}{\partial r} v + \frac{\partial v}{\partial t}$$

Substituting value in eqn (2) we get

$$F = \rho dA ds \left(\frac{\partial v}{\partial t} v + \frac{\partial v}{\partial t} \right) \quad \text{--- (3)}$$

Also,

$$\cos \theta = -\frac{\partial h}{\partial x}$$

Where h = verticle height of the tube from a given verticle plain-

Now, relation (1) becomes

$$F = -\frac{dp}{dx} \cdot dA da + mg \cos \theta$$

$$dA ds \left(\frac{\partial v}{\partial t} v + \frac{\partial v}{\partial t} \right) = -\frac{dp}{dx} \cdot dA da - mg \frac{\partial h}{\partial x}$$

$$dA ds \left(\frac{\partial v}{\partial t} v + \frac{\partial v}{\partial t} \right) = -\frac{dp}{dx} \cdot dA dx - dA ds g \frac{\partial h}{\partial x}$$

$$s \left(\frac{\partial v}{\partial t} v + \frac{\partial v}{\partial t} \right) = -\frac{dp}{dx} - g \frac{\partial h}{\partial x} \quad \text{--- (4)}$$

This eqn known as Euler's equation applicable for both steady and unsteady fluid flow. In the case of streamline flow velocity $\frac{dv}{dt} = 0$.

∴ Equation (4) becomes

$$\rho v dv = -dp - \rho g dh$$

$$\Rightarrow \rho v dv + dp + \rho g dh = 0 \quad \text{--- (5)}$$

$$\Rightarrow v dv + \frac{dp}{\rho} + g dh = 0$$

Integrating above equation along stream line we have

$$\frac{v^2}{2} + \int \frac{dp}{\rho} + gh = \text{constant} \quad \text{--- (6)}$$

This is called Bernoulli's equation. In the case of incompressible fluid

i.e. $\rho = \text{constant}$ above equation

$$\frac{v^2}{2} + \frac{p}{\rho} + gh = \text{constant} \quad \text{--- (7)}$$

If we consider unit volume of fluid i.e. mass = ρ of the fluid we have on multiplying eqn (7)

by ρ

$$\frac{1}{2} \rho v^2 + p + \rho gh = \text{constant}$$

This is another form of Bernoulli's eqn in terms of pressure

- ① Two drops of water of same size are falling through air with terminal velocity 1 m/s. If the two drops combine to form a single drop. calculate terminal velocity.
- ② A pipe is running full of water. At a certain point A it tapers from 0.6 m diameter to 0.2 m diameter at B. The pressure difference between A and B is 1 m of water column. Find the rate of flow of water through pipe.

- ③ Calculate the mass of water flowings in 10 minute through a tube 0.1cm in diameter, 40cm long if there is a constant pressure head of 20 cm of water. The coefficient of viscosity of water is 0.0089 cgs unit.

1 Solution:

Volume of a single drop = 2x volume of two drop

$$\frac{4}{3} \pi R^3 = 2 \times \frac{4}{3} \pi r^3$$

$$R^3 = 2r^3$$

R

$$\frac{4}{3} \pi R^3 = 2 \times \frac{4}{3} \pi r^3$$

$$R^3 = 2r^3$$

$$\therefore R = 2^{1/3} r \therefore r =$$

If V_1 be the terminal velocity of each drop and V_2 be combined drop velocity

Now, from Stokes law,

$$V_1 = \frac{2}{9} \frac{R^2 (\rho - \sigma) g}{\eta} \quad \text{--- (1)}$$

$$V_2 = \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta} \quad \text{--- (2)}$$

Dividing (2) by (1)

$$\frac{V_2}{V_1} = \frac{r^2}{R^2}$$

$$\frac{V_2}{V_1} = \frac{2^{2/3} r^2}{R^2}$$

$$\therefore \frac{V_2}{V_1} = 2^{2/3} V_1 \therefore 1.588 \text{ m/s}$$

2. Solution

Let P_1, v_1 be pressure and velocity at A and P_2, v_2 be pressure and velocity at B then

$$\text{Pressure difference } (P_1 - P_2) = 1 \times 10^3 \times 9.8 \text{ Nm}^{-2}$$

$$\text{Rate flow at A} = v_1 \times \text{area} = \pi \times 0.30^2 \times v_1 = 0.09 \pi v_1$$

$$\text{" " " B} = v_2 \times \text{area} = \pi \times 0.1^2 \times v_2 = 0.01 \pi v_2$$

For steady flow we have

$$0.09 \pi v_1 = 0.01 \pi v_2$$

$$\text{Rate flow at A} = \text{Rate flow at B}$$

$$0.09 \pi v_1 = 0.01 \pi v_2$$

$$v_2 = 9v_1$$

Height is constant then from Bernoulli's theorem

$$\frac{P_1}{\rho} + \frac{v_1^2}{2} = \frac{P_2}{\rho} + \frac{v_2^2}{2}$$

$$\frac{P_1 - P_2}{\rho} = \frac{1}{2} (v_2^2 - v_1^2)$$

$$\frac{1 \times 10^3 \times 9.8}{1 \times 10^3} = \frac{1}{2} (81v_1^2 - v_1^2)$$

$$\therefore v_1 = 0.495 \text{ m/s}$$

$$\therefore \text{Rate flow} = \pi \times 0.09 \times 0.495$$

$$= 0.14 \text{ m}^3/\text{sec}$$

3. Solution:

$$\text{length } l = 40 \text{ cm} \\ = 0.4 \text{ m}$$

$$\eta = 0.0089 \text{ C.G.S unit} \\ = 0.00089 \text{ decapoise}$$

$$p = 20 \text{ cm of water eqn. } [1 \text{ m} = 100 \text{ cm}]$$

$$= \frac{20}{100} \times 9.8 \times 10^3$$

$$= 1.96 \times 10^3 \text{ N/m}^2 = 1960 \text{ N/m}^2$$

$$\text{Rate flow } V = \frac{p \pi a^4}{8 \eta l}$$

$$\therefore \text{Radius } (a) = 0.05 \times 10^{-2} \text{ m} = 5 \times 10^{-4} \text{ m}$$

$$\text{diameter } (d) = \frac{0.1 \text{ cm}}{100}$$

Then,

$$V = \frac{1.96 \times 10^3 \times \pi \times (0.05 \times 10^{-2})^4}{8 \times 0.00089 \times 0.4}$$

$$\therefore V = 135.13 \times 10^{-9} \text{ m}^3$$

$$\text{Mass flowing / sec} = 135.13 \times 10^{-9} \times 10^3 \\ = 135.13 \times 10^{-6} \text{ Kg}$$

$$\text{Mass flow in 10 min} \\ = 135.13 \times 10^{-6} \times 600 \\ = 81.078 \times 10^{-3} \text{ Kg}$$

Thus for quarter wave plate if 't' be the thickness, μ_o and μ_e be the refractive index of 'o' and 'e' ray. Then path difference $= (\mu_o - \mu_e)t$
 $\Rightarrow \frac{\lambda}{4} = (\mu_o - \mu_e)t$

$$\therefore t = \frac{\lambda}{4(\mu_o - \mu_e)} \quad [\text{For-recrystall}]$$

Also,

$$t = \frac{\lambda}{4(\mu_e - \mu_o)} \quad [\text{For+ve crystal}]$$

It converts plane polarized light into circularly or electrically polarized light.

Half Wave Plate

It is a thin plane of birefringent crystal with optic axis parallel to its refracting face. When plane polarized light is incident on it,

Types Of Polarization

> Plane Polarized Light

It is a wave in which the oscillation occurs in a single plane. In a plane polarized light the oscillation of electric field vector $\vec{E}$ are strictly confined to a single plane perpendicular to the direction of propagation. As the direction of field vector at some point in space and time lies along a line perpendicular to direction of wave propagation a plane polarized wave also known as linearly polarized wave. With linear polarization the orientation of $\vec{E}$ vector space is constant at a point in space.

> Circularly Polarized Light

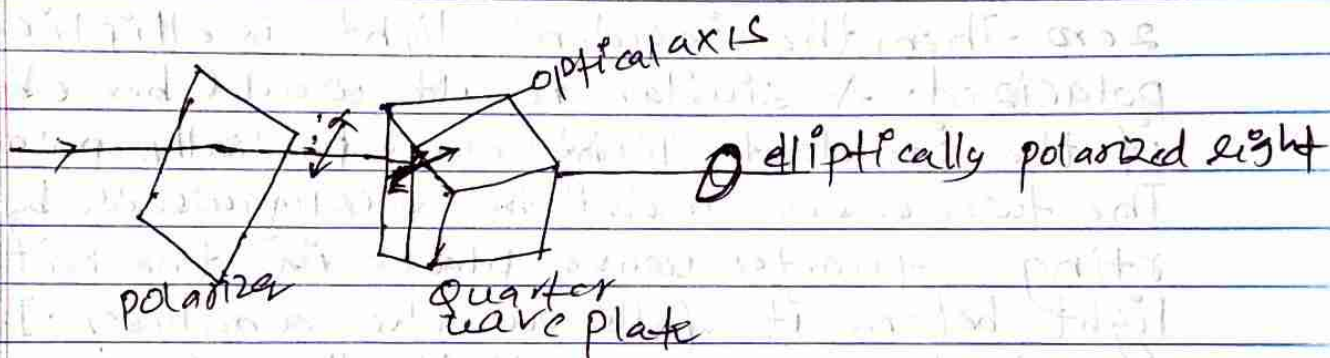
A light wave is said to be circularly polarized if in the course of wave propagation the magnitude of electric vector $\vec{E}$ stays constant but it rotates at a constant rate about the direction of propagation and sweeps a circular helix in space. If we could see the wave advancing towards our eyes we would find that the tip of $\vec{E}$ vector tracing a circle in space completing 1 revolution in within one (1) wavelength. Hence the state of polarization called circular polarization.

> Elliptically Polarized Light

A light wave said to be elliptically polarized, if the magnitude of $\vec{E}$ changes with time and vector $\vec{E}$

rotates about the direction of propagation and sweeps a flattened helix in space. If we imagine that we are looking at the light wave advancing towards us we would observe that the tip of the $\vec{E}$ traces an ellipse in space. Hence called electrically polarized light.

Production Of Elliptically Polarized Light



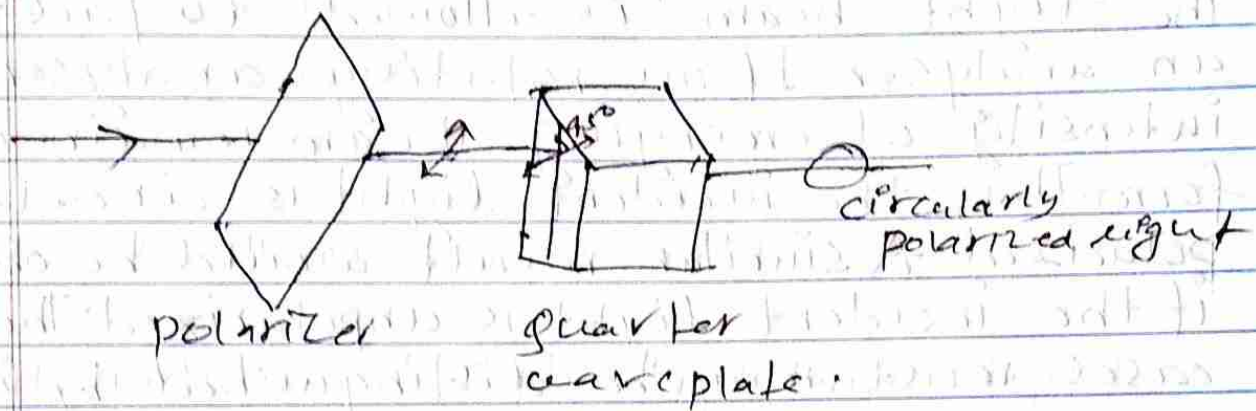
unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer as shown in figure. The plane polarized light is then made incident on a quarter wave plate. The quarter wave plate or the polarizer is rotated such that the electric vector $\vec{E}$ of plane polarized light makes an angle θ ($\theta \neq 45^\circ$) with optical axis of quarter wave plate. The incident ray divides into o and e-ray of amplitude $E \sin \theta$ and $E \cos \theta$. They travel along same direction in the crystal with different velocities. The two rays are polarized in orthogonal planes. They are in phase at front face but progressively get out of phase as they travel through crystal. When they emerge out of crystal they will have a

path difference of $\frac{\lambda}{4}$ and phase difference of $\frac{\pi}{2}$. When they combined produce elliptically polarized light.

Detection Of Elliptically Polarized Light

The light beam is allowed to pass through an analyser. If on rotating analyser. The intensity of emerging beam varies from a maxima to a minimum value but never reaching zero. Then, the incident light is elliptically polarized. A similar result would be obtained if the incident light were partially polarized. The two cases must be distinguished by inserting quarter wave plate in the path of a light before it falls on the analyser. If the original light is elliptically polarized, it may be considered as a resultant of two coherent plane polarized wave i.e. e-ray and o-ray which are out of phase by 90° . If the light passes through the quarter wave plate and additional phase difference of 90° is introduced between e and o-ray. Therefore the total phase difference becomes π between e and o-ray. On emerging from a quarter plate the e-ray and o-ray combined to produce plane polarized light. If the light coming out from quarter wave plate is examined with analyser, light will be extinguished twice in one full rotation of analyser.

Production Of Circularly Polarized Light



unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer. Plane polarized light is then made incident on a quarter wave plate. The polarizer and quarter wave plate are rotated such that the electric vector $\vec{E}$ of the plane waves makes an angle of 45° with optic axis of quarter wave plate. The plane polarized wave incident on quarter wave plate split into two rays travel in the same direction inside the crystal but with different velocities. The two rays are in phase at front face of crystal but progressively get out of phase as they travel through the crystal. As they emerge from rear face of the crystal they will have a path difference of $\frac{1}{4}$ or phase difference of 90° . The two rays are linearly polarized in mutually $\perp$ direction. When they combine they produce circularly polarized light.

Detection of circularly Polarized Light

The light beam is allowed to pass through an analyser. If on rotating analyser the intensity of emerging beam remains uniform. Then, the incident light is circularly polarized. A similar result would be obtained if the incident light is unpolarized. The two cases must may be distinguished by inserting a quarter wave plate in the path of light before it falls on analyser. If original light is circularly polarized it may be considered as a resultant of two coherent plane polarized wave i.e. e-ray and o-ray which are out of phase by 90° . If the light passes through quarter wave plate an additional phase difference of 90° is introduced between e-ray and o-ray. Therefore total phase difference becomes 180° between e-ray and o-ray. On emerging from quarter wave plate the e and o-ray combined to produce plane polarized light. Therefore if light coming out from quarter wave plate is examined with an analyser light will be extinguished twice in one full rotation of the analyser.

Malus Law

The amount of light transmitted through a analyser at an arbitrary angle is given by Malus law.

It state that, "the intensity of polarized light transmitted through an analyser is proportional

to the square of cosine of angle between the plane of polarization of light and the transmission axis of polarizer.

If unpolarized light of intensity I_0 is incident on a polarizer. Plane polarized light of intensity $I_0/2$ is transmitted by it. Let this plane polarized light pass through an analyzer. The intensity of light transmitted through analyzer given by

$$I = \frac{I_0}{2} \cos^2 \theta$$

Specific Rotation

Specific rotation of optically active substance is defined as the rotation produced by a decimeter (1 decimeter = 10 cm) long column of the liquid containing 1 gm of the active substance 1 cc of the solution.

$$[S]_D^T = \frac{\alpha}{l \cdot c}$$

$\alpha \rightarrow$ (Angle of rotation)

Where,

$[S]_D^T$ = Specific rotation at temperature $T^\circ\text{C}$ for a wave length λ

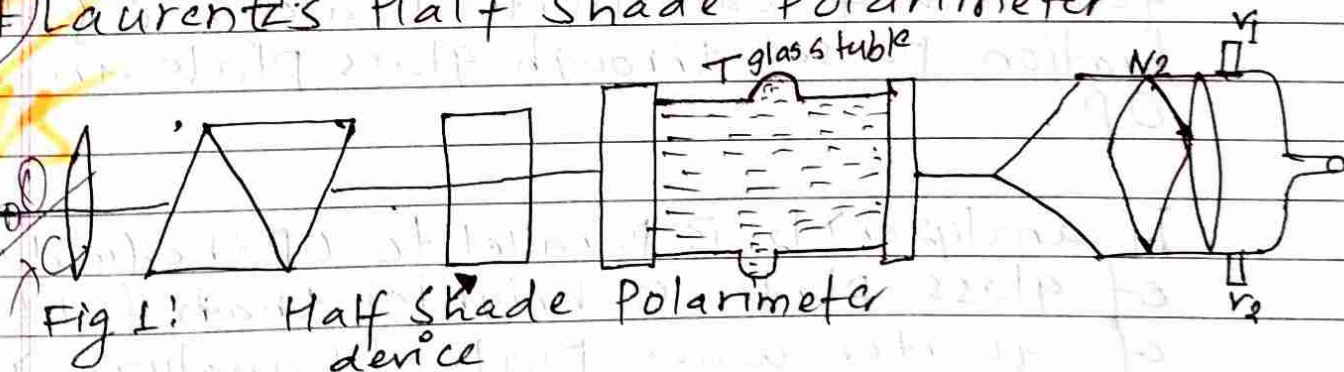
α = Angle of rotation

l = length of solution in centimeter through which plane polarized light passes (cm)

c = concentration of the active substance in gm/cc



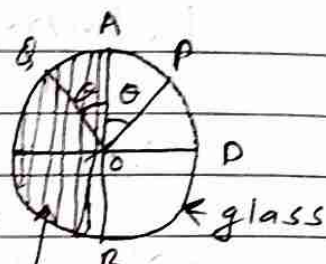
Laurent's Half Shade Polarimeter



Polarimeter is an optical device used for determination of specific rotation of solution. It consists of two Nicol prism N_1 (Polarizer), N_2 (Analyzer) and half shade plate. The half shade plate consists of semicircular quartz half wave plate and a semi-glass plate. Quartz-half wave plate

introduces a phase difference of π between two components e-ray and o-ray passing through it and thickness of glass plate is chosen in such a way that it transmit same amount of light as done by quartz plate.

Consider the incident vibration OP makes an angle θ with the optic axis which is parallel to the line of separation AB. The



half vibration which passes through glass plate passes as such but the vibration incident on half wave plate split into o-wave and e-wave. O-waves are along OD and e-waves along OA since half wave plate introduces phase difference of π and o-ray emerges out of plate along OC instead of OD. Hence, resultant vibration passes through quartz half wave plate is along OQ, but vibration passes through glass plate are along OP.

If analyzer N_2 is parallel to OP the field of view of glass plate is brighter than field of view of quarter wave plate. If analyzer N_2 is parallel to OQ the quartz half will be brighter than glass half. If is parallel to CD both half are equally dark. Now, the analyzer

Now, analyzer is set in the position of equal bright without solution in glass tube. The circular scale reading is taken. Now,

glass tube is filled with optically active substance of known concentration the field of view is partially illuminated. So, the analyzer is rotated to get again the position of equally bright condition and reading is again taken. The difference in circular scale reading gives the angle of rotation of plane of vibration of plane polarized light. Knowing value of θ and c specific rotation of optically active substance can be determined by relation

$$[\alpha] = \frac{\theta}{lc}$$

where θ = is in degree

l = length of soln in (cm)

c = concentration is gm/cc

① Calculate thickness of half wave plate of quartz for a wave length of $5000\text{\AA}$.

Given $\mu_e = 1.553$ and $\mu_o = 1.544$

② Plane Polarized light passes through quartz plate with its optic axis parallel to the faces. Calculate least thickness of plate for which emergent beam will be circularly polarized. (Given $\mu_e = 1.5533$ and $\mu_o = 1.5442$)